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Sketch-to-Energy model automation with multimodal LLMs and retrieval-augmented generation☆

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ABSTRACT

Building Energy Modeling (BEM) is critical for improving building energy efficiency. However, modeling in the early architectural design stages is often hindered by challenges such as extracting geometric information from sketches and configuring complex parameters. To address these challenges and enable efficient energy analysis in the early design phases, we propose an automated BEM method. Users only need to provide preliminary architectural floor plan sketches and modeling requirements in natural language, after which our method can automatically generate complete simulation models. The proposed method features two major innovations. First, we leverage multimodal large language models (multimodal LLMs) and computer vision techniques to generate structured 3D building geometry from hand-drawn sketches. The multimodal LLM denoises errors in sketches and extracts room functions and dimensions, while computer vision methods generate wall and window coordinates. Second, we employ a retrieval-augmented generation (RAG) technique to retrieve similar cases within the same ASHRAE climate zone from an EnergyPlus model database. The BEM parameters of these similar cases are used as prompts to guide the LLM in generating thermal parameters and operational schedules that align with real-world scenarios. The effectiveness of the method was validated through four test cases with diverse morphologies across three ASHRAE climate zones. Quantitative results demonstrate robust geometric reconstruction capabilities, achieving a mean wall length error of 0.05 m and a room function classification accuracy of 91.2%. The thermal and operational parameters generated are empirically grounded, resulting in simulation models that exhibit high consistency with expert-built benchmarks (average $R^2 > 0.995$ for annual hourly loads). The average total modeling time was 8 min. Overall, the method provides an automated bridge between early architectural design and energy performance simulation in the demonstrated cases, enabling rapid conversion of conceptual sketches into executable models while supporting data-driven design optimization through customizable parameter retrieval weights.

1. Introduction

1.1. Importance and challenges of building energy modeling

Amid global efforts to address climate change and the energy crisis, the building sector, as a major contributor to energy consumption and carbon emissions, faces an critical need to enhance energy efficiency [1]. Building Energy Modeling (BEM) serves as a critical technology for evaluating and optimizing building energy performance, playing an essential role in demand-side management [2], HVAC system optimization, and building design operation [3]. To facilitate performance-driven design, BEM needs to be integrated into the early stages of the building design process to support rapid performance assessment and iterative optimization of design schemes [4,5]. Currently, simulation engines such as EnergyPlus [6,7], TRNSYS [8], DeST [9,10] and Dymola [11] are widely used due to their good interpretability and modular structure. However, traditional BEM workflows are hindered by three major challenges.

1) **Complex parameter configuration** [12,13]. Developing high-quality energy models heavily relies on expert knowledge and extensive manual configuration [14]. The traditional BEM process involves multiple complex steps [15], including geometric modeling, envelope

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parameter settings, and operational schedule definition. Iterative adjustments in these steps are time-consuming and susceptible to human errors [16].

2) **Difficulty in processing unstructured inputs** [17]. In the early design phase, designers typically provide architectural sketches and textual requirement specifications, which are inherently unstructured. Although these sketches contain rich geometric and semantic information, including walls, room layouts, and key dimensions, they are often filled with errors and noise. Converting these messy sketches into structured data (e.g., EnergyPlus IDF files) represents a significant bottleneck in traditional BEM workflows. Traditional rule-based or vision-based methods struggle to process such inputs effectively, hindering timely energy efficiency feedback during critical early design stages and limiting performance-driven design optimization.

3) **Underutilization of historical BEM models**. The architecture industry has accumulated numerous high-quality historical building energy models (BEMs), typically stored as EnergyPlus input data files (IDFs) [18]. These calibrated and validated models encapsulate valuable design and optimization knowledge but are often archived and underutilized after project completion [19]. Given the high consistency in energy characteristics among similar building types, systematically transferring knowledge from historical models to new projects could significantly enhance modeling efficiency [20]. However, current practices rely heavily on manual retrieval, leading to substantial waste of valuable digital assets [21].

By automatically converting design sketches and textual requirements into BEM inputs, designers can evaluate multiple design alternatives to optimize energy efficiency in early stage [22]. This automated BEM workflow not only significantly reduces modeling time and minimizes human errors but also enhances overall project quality. Consequently, there is an critical need for an intelligent BEM workflow that enables efficient conversion from conceptual sketches to simulation −ready models, thereby improving the efficiency and accessibility of building energy analysis.

1.2. Potential of large language model and retrieval-augmented generation in BEM

The rise of large language models (LLMs) offers a promising pathway for streamlining the BEM workflow. LLMs, such as BERT [23], GPT [24], PaLM [25], DeepSeek [26], LLaMA [27] and Gemini [28], are large-scale models trained on vast text corpora [29]. They exhibit remarkable performance in natural language understanding [30], few-shot learning [31], and multi-step reasoning [32], achieving significant results in fields like healthcare [33–35], education [36–39], and law [40,41]. Since EnergyPlus IDF files are essentially text files with specific syntax, LLMs hold the potential to automate modeling by parsing natural language requirements and generating syntactically correct simulation input files. However, the complexity of BEM automation surpasses general text tasks, and directly applying pre-trained LLMs to BEM faces challenges [42].

The BEM often originates from architecture sketches or simple floor plans. Multimodal LLMs (e.g., GPT-4o [43], Gemini-2.5-Pro [28]) demonstrate strong capabilities in general image understanding, scene reasoning, and optical character recognition (OCR) tasks, enabling the simultaneous processing of visual and textual information [44]. This capability provides a new pathway for the collaborative extraction of geometric and semantic information from sketches. In the field of mechanical design [45], the CAD-GPT model proposed by i-WiN team [46] exemplifies this potential by mapping 2D sketch parameters to a language feature space to guide precise 3D model generation. Additionally, specialized computer vision models, such as the convolutional neural network (CNN)-based DeepFloorplan [47], can extract key geometric elements through pixel-level segmentation of floor plans, compensating for potential limitations of general multimodal LLMs in accurately localizing building components. Together, these technologies offer a feasible approach to bridge the critical gap between preliminary design intent and structured BEMs geometric information.

Meanwhile, BEM parameter configuration involves diverse thermal properties and operational schedules that must adhere to physical constraints and domain knowledge, which pre-trained models alone struggle to meet with sufficient accuracy [16]. While model fine-tuning [48,49] enhances performance by training pre-trained models with domain-specific data, it demands large high-quality datasets and substantial computational resources (e.g., high-performance GPUs/TPUs), making it costly. In contrast, prompt engineering [42] guides pre-trained LLMs to generate accurate responses in specific domains through carefully designed prompts [50], offering a more cost-effective and flexible approach [51]. Common prompt strategies include Chain of Thought (CoT) [52], Reasoning and Acting (ReAct) [53], and Retrieval-Augmented Generation (RAG) [54]. Among these, RAG enhances LLM outputs by dynamically retrieving relevant information from an external knowledge base to serve as context. This method not only mitigates the risk of generating factually or physically inaccurate hallucinations [55] but also allows model capabilities to be updated by simply enriching the knowledge base [56]. In the BEM context, RAG can leverage historical IDFs model libraries to provide LLMs with relevant physical constraints and operational patterns, effectively guiding the generation of compliant parameters. This constitutes a more scalable and adaptable solution compared to fine-tuning.

Recent studies have begun exploring LLMs in the building energy domain, focusing on three main directions: **1) Data analysis and interpretation.** Zhang et al. [57] developed a GPT-based method to mine and analyze operational data from building chiller systems, demonstrating LLMs' ability to handle complex operational conditions. Rysanek et al. [58] further validated LLMs' potential in reasoning with building science data, generating credible analysis results in academic language. **2) System control and operation.** Song et al. [59] utilized GPT-4 to optimize HVAC system control, demonstrating LLMs' potential in real-time operational decision-making. 3) **Energy model generation.** Directly relevant to this study, Jiang et al. [60] built the Eplus-LLM platform by fine-tuning LLMs to generate building energy models. Forth and Borrmann [61] improved LLMs' accuracy in handling text semantic similarity tasks in the building domain through fine-tuning multilingual models, enhancing the semantic richness of Building Information Modeling (BIM). Zhang et al. [62] proposed a multi-agent LLMs approach to optimize the modeling process. Despite demonstrating LLMs' potential, existing research on model generation has limitations. The methods by Jiang et al. [60] and Forth et al. [61] rely on fine-tuning, requiring large domain-specific datasets and high computational resources. Although Zhang et al. [62] employed a multi-agent architecture to guide LLM responses through prompts, their approach did not incorporate RAG technology. Users are still need to provide detailed envelope structures, materials, and internal load information (e.g., wall layering details). If a user can only supply simple parameters like wall heat transfer coefficients and cannot provide complete inputs, the modeling process is hindered.

Notably, existing methods primarily focus on processing structured or textual inputs, failing to leverage multimodal LLMs or advanced computer vision technologies to address the challenge of generating structured 3D building geometric models from architecture sketches. Furthermore, these studies have not effectively utilized the rich engineering experience embedded in historical BEMs through techniques like RAG, missing opportunities to reuse valuable digital assets and ensure parameter accuracy. How to effectively combine the semantic-visual reasoning capabilities of multimodal LLMs with building domain knowledge to achieve automated generation from sketches to simulation-grade geometric models remains an research gap.

1.3. Aims and approach

To address these challenges, this study proposes an automated BEM

method for generating simulation-ready EnergyPlus models from architecture sketches and modeling requirements in natural language. The proposed method uses multimodal LLMs and computer vision technologies to construct EnergyPlus geometric models from architecture sketches, and simultaneously employs RAG technology to ensure the accuracy of modeling parameters by efficiently reusing expert knowledge from historical BEMs. The core objective of this method is to automate the process from early, unstructured design information (i.e., sketches and simple textual descriptions) to the generation of simulation-grade EnergyPlus models, thereby significantly reducing technical barriers and resource demands. To achieve this, the proposed method systematically decomposes the modeling process into five core tasks: Requirements Parser, Building Geometry Generator, Thermal Parameter Generator, Operational Parameter Generator, and IDF Generator and Simulator. Through its modular design and multi-technology synergy, the method enhances modeling efficiency while ensuring simulation reliability, lowering the technical threshold and resource requirements for BEM and promoting the widespread adoption and application of building energy analysis in the early stages of design.

The contributions of this study include: 1) Proposing an automated BEM method, which supports the generation of complete and simulatable EnergyPlus models from unstructured multimodal inputs, i.e., hand-drawn sketches and textual descriptions); 2) Developing a geometric reconstruction method that leverages multimodal LLMs and computer vision to generate structured 3D building geometry from unstructured, noisy sketches during the early design phase; 3) Introducing the RAG mechanism into the BEM parameter generation process, leveraging expert knowledge and physical constraints from historical BEMs to significantly enhance the accuracy and physical plausibility of generated parameters.

The structure of this paper is organized as follows: Section 2 provides a detailed introduction to the proposed method, including its workflow, the construction method of the EnergyPlus model database, and the implementation details of each task module; Section 3 validates the effectiveness of the method through case studies; Section 4 discusses the advantages and limitations of this study and outlines future research directions; Section 5 presents the conclusions.

2. Methodology

This section details the proposed automated BEM framework. Section 2.1 provides a comprehensive overview of the overall workflow. Section 2.2 describes the construction of the EnergyPlus model database, which serves as the knowledge base. Finally, Section 2.3 presents the in-depth implementation details for each core task module.

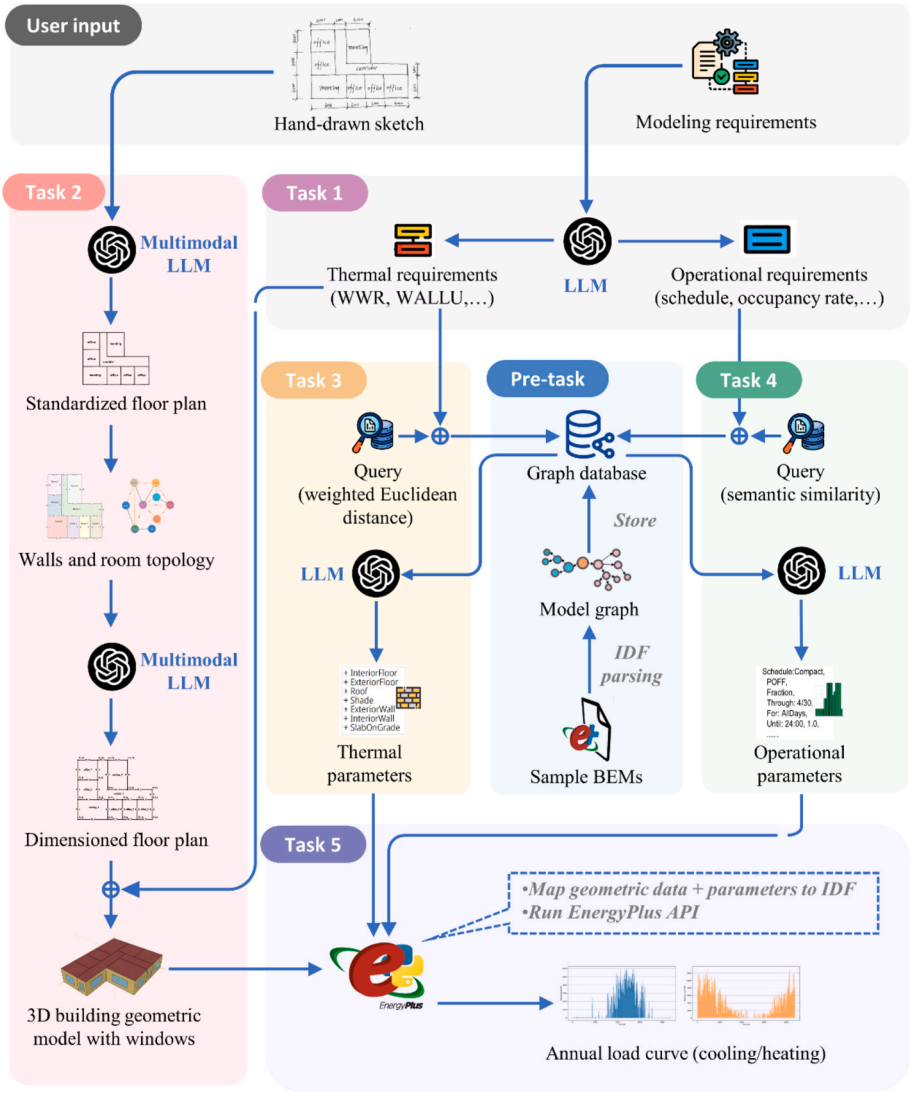


Fig. 1. Overall workflow of the proposed method.

2.1. Overview of the proposed automated BEM method

As illustrated in Fig. 1, the proposed method optimizes the traditional BEM workflow by integrating multimodal LLMs, computer vision techniques, and RAG technology. The method accepts two types of inputs: 1) unstructured textual modeling requirements, where users describe key building features, such as geometric characteristics, thermal properties, spatial functions, and operational parameters, in natural language; and 2) architecture sketches, provided as hand-drawn or scanned images. These visual inputs, which encapsulate preliminary design intent (e.g., wall layouts, room divisions, and dimensional annotations), are unstructured and processed automatically to generate structured building geometry models compatible with EnergyPlus simulations.

The decision to incorporate architecture sketches as a primary input is driven by several considerations. First, for complex spatial layouts, providing a fully detailed and unambiguous textual description is labor-intensive and challenging. Second, natural language inherently struggles to convey precise geometric relationships, such as adjacency, alignment, or room partitioning. Finally, compared to general-purpose LLMs, which are less adept at complex spatial reasoning, multimodal LLMs combined with computer vision techniques can more robustly extract structured geometric and topological relationships from sketches.

The automated BEM method is decomposed into five core task modules, which work in concert to manage the entire process from raw input to final simulation.

1) **Task 1: Requirements Parser**: This module utilizes a structured prompt to guide an LLM in parsing the user's natural language input. It accurately extracts key geometric, thermal performance, and operational parameters, and outputs them as standardized JSON-formatted data.
2) **Task 2: Building Geometry Generator**: This module is responsible for generating structured 3D building geometric model from the hand-drawn sketch. It first employs a multimodal LLM to denoise errors in original sketch. Subsequently, it employs computer vision techniques to extract wall pixel coordinates and room topology recognition. A multimodal LLM is then used again to extract semantic information such as room functions and dimensions. By solving a system of linear equations, it infers unannotated wall dimensions, achieving a precise mapping from pixel coordinates to physical planar coordinates. Finally, based on the layer height and window-to-wall ratio (WWR) parsed in Task 1, a complete 3D model that includes windows is generated.
3) **Task 3: Thermal Parameter Generator**: This module employs the RAG technique. Based on the thermal characteristics extracted in Task 1, it retrieves similar cases from the EnergyPlus model database using a weighted Euclidean distance metric. The detailed constructions of these cases are then used as context to guide the LLM in generating a complete set of thermal parameters that align with the design objectives.
4) **Task 4: Operational Parameter Generator**: This module also uses the RAG technique but focuses on the textual operational information parsed in Task 1. It retrieves the operational schedules of similar cases based on semantic similarity and uses them as context to guide the LLM in generating detailed operational parameters covering all required time periods.
5) **Task 5: IDF Generator and Simulator**: As the final stage, this module integrates and maps the structured data generated from Task 2 (geometry), Task 3 (thermal), and Task 4 (operational). It automatically compiles this information to generate an EnergyPlus Input Data File (IDF), calls an API to execute the simulation, and outputs results such as hourly load curves.

2.2. Construction of the EnergyPlus model database (pre-task)

The EnergyPlus model database serves as the knowledge base for the RAG framework, storing high-quality commercial reference building models (BEMs). To enable efficient retrieval and generation, raw IDF files are parsed into a structured graph schema consisting of two tiers: **Key Features** and **Detailed Information**. Key Features comprise high-level parameters (e.g., Area, WWR, U-values) that significantly impact energy consumption (Table 1). The selection of these features follows two main principles. First, they are widely recognized in academic research as core energy performance indicators [63–65]; Second, their level of abstraction aligns with the limited information available in the early design stages, allowing them to serve as a basis for rapid and robust similarity matching [66]. Detailed Information contains granular model data, specifically envelope constructions (e.g., wall layers and material properties) and operational schedules. The Key Features and Detailed Information are not directly used in the initial similarity retrieval but serves as the contextual knowledge injected into the LLM prompt after a match is found. It provides validated, high-quality reference information for Task 3 (Thermal Parameter Generator) and Task 4 (Operational Parameter Generator), which is crucial for mitigating model hallucinations and generating professional-grade parameters.

As shown in Fig. 2, the model database is built using the U.S. Department of Energy (DOE) commercial reference building models as its data source. All IDF files first undergo data cleaning and standardization to remove incomplete data and unify physical units. Subsequently, the IDFs are parsed according to the aforementioned data schema to extract Key Features and Detailed Information. Finally, through graph construction, a Key Features Graph and a Detail Information Graph are generated and integrated into a Model Graph, which is stored in a Neo4j graph database. A graph database was chosen because it can efficiently represent the complex hierarchical and connective relationships between building components (e.g., the subordinate relationship between materials and constructions). Furthermore, its query language and algorithms support the complex similarity retrieval and semantic matching required by the method, while also offering the dynamic scalability needed for future expansion of the model database.

2.3. Task details

2.3.1. Requirements parser (task 1)

To address the diversity and ambiguity of natural language inputs, this task employs a structured prompt based on in-context learning (ICL). The prompt systematically guides the LLM by predefining an expert role (Role), specifying task objectives (Task description), giving detailed instructions (Instructions), and providing examples (Example). This structured approach, particularly through the inclusion of synonym lists and clear parameter definitions, significantly enhances the model's ability to handle varied expressions, ensuring the accuracy and consistency of extracted results.

The implementation details are exemplified by specialized prompt templates: one for extracting geometric and thermal requirements (Fig. 3) and another for parsing operational information (Fig. 4). Fig. 5 presents a complete example of extracting and structuring various modeling requirements from a typical user input. For geometric and thermal requirements, Task 1 extracts the key features defined in Table 1, standardizing the units of extracted values to comply with EnergyPlus modeling conventions (e.g., unifying heat transfer coefficients to $W/(m^2 \cdot K)$) for direct use in Tasks 3 and 4. For operational requirements, it extracts information such as room types, occupancy rates, lighting power density, and schedules, initially generating schedules in JSON format. Note that while operational schedules are initially parsed here, their detailed refinement is delegated to Task 4 via retrieval augmentation.

Table 1
Key features of building energy models and selection reasons.

Category	Feature name	Abbreviation	Unit	Selection reasons
Building location	Building location	LOCATION	–	• Climatic conditions (e.g., temperature, humidity, and solar radiation) in different regions influence the cooling/heating demands
Building geometry	Building area	AREA	m^2	• Usage space and vertical structure affect the internal thermal environment and HVAC load distribution.
	Number of layers	NL	–	• Lower CR typically indicates reduced energy loss.
	Layer Height	H	m	• WWR directly affects natural lighting and thermal load; extreme window areas can significantly alter energy consumption.
	Compactness ratio	CR	–	
	Window-to-wall ratio	WWR	–	
Envelope thermal performance	External wall heat transfer coefficient	WALLU	W/(m^2·K)	• The thermal conductivity determines the insulation performance of the envelope; lower U-values help reduce heat loss and improve energy efficiency.
	External wall heat capacity	WALLSP	kJ/(m^2·K)	
	Roof heat transfer coefficient	ROOFU	W/(m^2·K)	
	Window glass heat transfer coefficient	WINDOWU	W/(m^2·K)	
	Slab on grade floor heat transfer coefficient	SLABU	W/(m^2·K)	

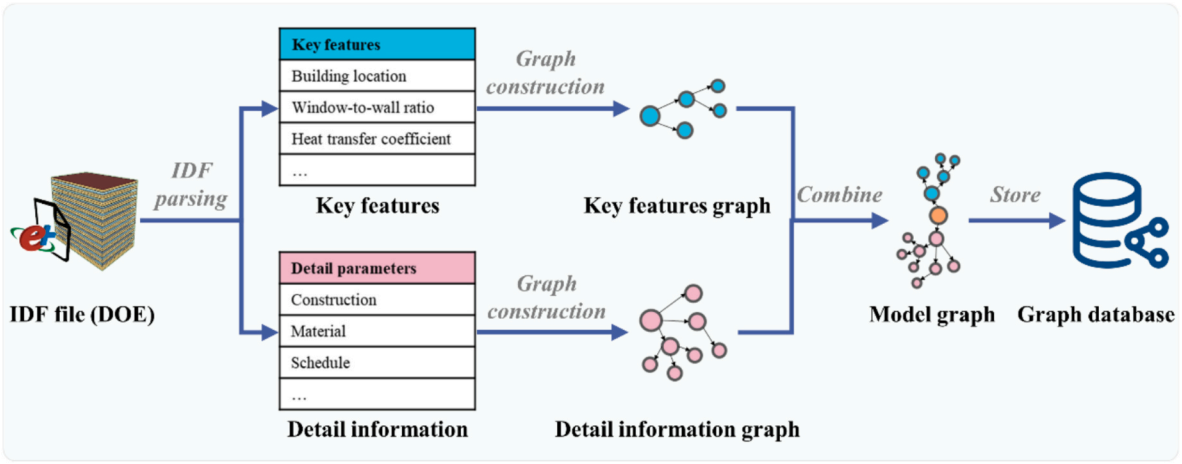


Fig. 2. EnergyPlus model database construction process.

2.3.2. Building geometry generator (task 2)

The geometric reconstruction process integrates multimodal LLMs with computer vision through the following steps:

1) Multimodal LLM-driven architecture sketch denoising

Hand-drawn sketches often contain noise such as irregular lines, inconsistent symbols, and distorted proportions. These uncertainties pose significant challenges to the automated and precise extraction of geometric information. Structured prompts are first used to guide GPT-4o in standardizing the input. As shown in Fig. 6, the prompt focuses the model's visual understanding on enclosure structures (walls) as spatial boundaries, adhering to a "wall-first" principle. Based on topological adjacency and geometric constraints, the model redraws the original sketch, effectively removing visual noise such as line jitter, breakpoints, and artifacts while preserving the core spatial layout and design intent. This process yields a topologically correct and geometrically regularized standardized floor plan (SFP). The resulting SFP provides high-quality input for subsequent wall vectorization, forming the foundation for the robustness of the automated workflow.

2) Wall coordinates extraction and room topology identification

To extract wall pixel coordinates and construct room topology, the CNN-based DeepFloorplan model is applied to segment the SFP and generate a binary wall mask image M_W, defined as:

$$M_W(x,y) = \begin{cases} 1 & if\ pixel\ (x,y) \in wall \\ 0 & otherwise \end{cases} \quad (1)$$

where (x,y) represents pixel coordinates. The generation of M_W provides a precise pixel foundation for subsequent wall coordinates extraction.

A skeletonization algorithm then reduces the wall regions to single-pixel-width centerlines from the binary mask M_W. To enhance line continuity and clarity, the image is preprocessed preprocessing is performed using Gaussian blur and morphological closing operations, followed by Canny edge detection to strengthen skeleton edges. Probabilistic Hough Transform is then applied to detect line segments in the skeleton image. By leveraging pixel-to-parameter space mapping and voting mechanisms, proposed method effectively mitigates image noise interference, yielding an initial set of wall line segments. To improve geometric accuracy and topological integrity, further optimization is applied to the wall line set. First, collinear merging is performed on nearly parallel and spatially adjacent segments, with directional similarity assessed via the cosine similarity of their respective direction vectors, v_1 and v_2. Segments are merged when $\cos\theta$ approaches 1 and the endpoint distance is below a preset threshold.

$$\cos\theta = \frac{v_1 \bullet v_2}{\|v_1\|\|v_2\|} \quad (2)$$

Second, to address gaps at wall corner, a wall line extension and closure mechanism is implemented. Based on endpoint neighborhood detection, the perpendicular distance d from a wall endpoint (x_0, y_0) to a nearby wall segment defined by points (x_1, y_1) and (x_2, y_2) is calculated; if d is below a threshold, an extension is executed.

Prompt template for thermal features extraction

Role

You are a helpful HVAC expert, specialized in building modeling.

Task description

Extract the key properties related to the building's thermal performance from the following text and return them in JSON format.

Context

{Modeling_requirements}

Instructions

- Extract all key properties. If a property is not mentioned, set its value to null.
- The key information to be extracted and its synonymous expressions are listed below. When parsing the text, the information related to each of the above key fields should be understood and extracted based on the context of the sentence to ensure accurate mapping to the corresponding JSON fields:
  - "location": "Location, which could be described as any of the following: location, place, site",
    ……
  - "NL": "Number of Layers,……

Example

Example Sentence:
"The structure in Shanghai has a form factor of 0.51, a window-to-wall ratio of 0.42, a wall heat transfer coefficient of 0.36 W/(m²·K), a wall heat retention of 150 J/(m²·K), a floor heat transfer coefficient of 0.45 W/(m²·K), a roof heat transfer coefficient of 0.50 W/(m²·K), a daylight opening u-value of 2.2 W/(m²·K), and a built-up area of 1400 m² over 12 floors."

Example Output:

```
{{
  "location": "Shanghai",
  "CR": "0.51",
  "WWR": "0.42",
  "WALLU": "0.36 W/(m²·K)",
  "WALLSP": "150 J/(m²·K)",
  "FLOORU": "0.45 W/(m²·K)",
  "ROOFU": "0.50 W/(m²·K)",
  "WINDOWU": "2.2 W/(m²·K)",
  "AREA": "1400 m²",
  "NL": "12",
}},
```

Fig. 3. Prompt template for thermal requirements extraction.

$$d = \frac{\left|(y_2-y_1)x_0 - (x_2-x_1)y_0 + x_2y_1 - y_2x_1\right|}{\sqrt{(y_2-y_1)^2 + (x_2-x_1)^2}} \tag{3}$$

To construct an accurate topology graph, all wall lines must be precisely segmented at intersection points. For any two line segments L_1, with endpoints $((x_1,y_1),(x_2,y_2))$, and L_2, with endpoints $((x_3,y_3),(x_4,y_4))$, the intersection point is derived by solving parametric equations. An intersection occurs only when $0 \le u_a \le 1$ and $0 \le u_b \le 1$, with the intersection point P calculated as per Equation (6).

$$u_a = \frac{(x_4-x_3)(y_1-y_3) - (y_4-y_3)(x_1-x_3)}{(y_4-y_3)(x_2-x_1) - (x_4-x_3)(y_2-y_1)} \tag{4}$$

$$u_b = \frac{(x_2-x_1)(y_1-y_3) - (y_2-y_1)(x_1-x_3)}{(y_4-y_3)(x_2-x_1) - (x_4-x_3)(y_2-y_1)} \tag{5}$$

$$P = (x_1 + u_a(x_2-x_1), y_1 + u_a(y_2-y_1)) \tag{6}$$

All detected intersection points serve as segmentation points, dividing the original wall lines into finer-grained independent segments. Through these steps, a precise set of wall line segments is obtained. This set is immediately mapped into a graph structure, with wall endpoints as nodes (Vertices) and wall segments as edges (Edges). Subsequently, a depth-first search (DFS) algorithm is employed, initiating from each node to systematically traverse all paths and detect closed loops (Cycles). To ensure the uniqueness of each room boundary, nested paths are eliminated based on geometric containment relationships, and duplicate paths are filtered out. The process ultimately outputs a set of closed paths containing wall segment information, where each path represents an independent room.

3) Semantic annotation and geometric dimension inference

Gemini-2.5-Pro is employed to perform Optical Character Recognition (OCR) and semantic reasoning on the original sketch, extracting room function labels (Fig. 7) and dimensional annotations (Fig. 8). For an example, as shown in Fig. 7, the model is guided to analyze the textual labels (e.g., "Office") within each identified room area through a specifically designed prompt. If textual labels are absent, the model infers the function based on visual cues such as furniture layout and spatial

Prompt template for operational load extraction

Role

You are a helpful HVAC expert, specialized in building modeling.

Task description

Extract the key properties related to room type, occupancy rate, lighting power density, equipment power density, and the schedule from the following text and return them in JSON format.

Context

{Modeling_requirements}

Instructions

- Extract the room type (e.g., Office, Meeting, Corridors).
- Extract all key properties related to occupancy, lighting, and equipment power, as well as the time schedule if mentioned.
- The schedule must cover a full 24-hour period (00:00 to 24:00).
- Do not assume or extend any occupancy values for unmentioned periods.

Example

Example Sentence:
The office lobby has a lighting power density of 9.00 W/m². Occupancy begins at 0% before 07:00, rises to 10% by 09:00, reaches 20% by 11:00, stays steady at 20% until 18:00, and decreases to 5% by 20:00, reaching 0% after 22:00.

Example Output:

```
{{
  "room_type": "lobby",
  "conditioning_setpoint_temperature": "null",
  "occupancy_rate": "null",
  "lighting_power_density": 9.00,
  "equipment_power_density": {{
    "summer": "null",
    "winter": "null"
    }},
  "schedule": {{
    "00:00-07:00": 0,
    "07:00-09:00": 10,
    "09:00-11:00": 20,
    "11:00-18:00": 20,
    "18:00-20:00": 5,
    "20:00-22:00": "null",
    "22:00-24:00": 0,
  }},
}}
```

Fig. 4. Prompt template for operational requirements extraction.

configuration.

Simultaneously, another prompt (as illustrated in Fig. 8) directs the model to extract explicit dimensional annotations and associate them with their corresponding wall IDs. This process culminates in the generation of structured JSON data, completing the initial semantic and dimensional annotation.

Before performing geometric length inference, the system must establish an absolute scale between pixel space and physical space. The algorithm iterates through all annotated walls, calculates the ratio of their physical length to their pixel Euclidean distance, and takes the average as the global scale. This approach of averaging multiple annotated points effectively mitigates measurement errors caused by stroke jitter or local scale distortions inherent in hand-drawn sketches.

To resolve unannotated wall lengths, we propose a length inference technique based on the geometric closure constraint of each room. This technique leverages the fundamental geometric principle that for any closed polygon, the sum of its vector projections onto an orthogonal coordinate system is zero. For each room, two linear constraint equations can be established, as shown in Equation (7):

$$\begin{cases} \sum_{i\in\mathscr{U}} s_i \bullet L_i \cos\theta_i + C_H = 0 \\ \sum_{i\in\mathscr{U}} s_i \bullet L_i \sin\theta_i + C_V = 0 \end{cases} \tag{7}$$

where $\mathscr{U}$ represent the sets of all unknown walls along the room boundary; L_i is the length of the i-th unknown wall to be solved; $s_i \in \{+1, -1\}$ and θ_i are the direction coefficient and inclination angle, respectively, extracted from the geometric topology recognition step. and C_H and C_V are constant terms calculated from the vector sums of all known wall segments within the room (i.e., those with explicit dimen-

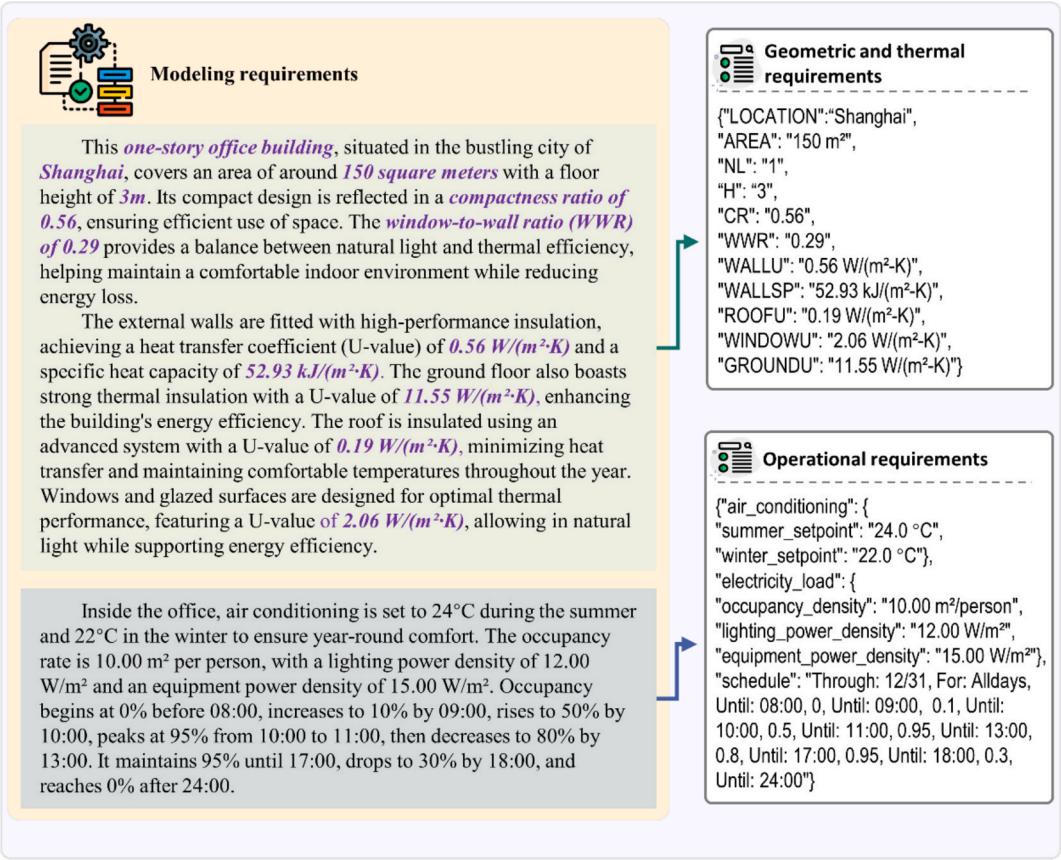


Fig. 5. Example of parsing thermal and operational requirements from textual modeling requirements.

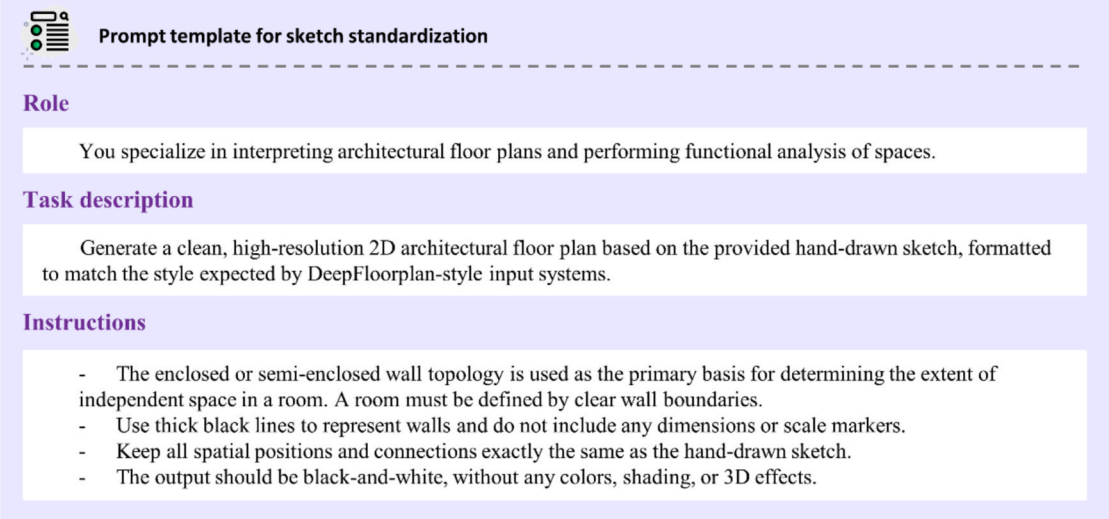


Fig. 6. Structured prompt template for sketch denoising.

sional annotations).

By aggregating the constraint equations from all rooms in the layer, a global sparse system of linear equations is formed:

$$\mathbf{A} \bullet \boldsymbol{L} = \mathbf{b} \tag{8}$$

where **A** is a sparse coefficient matrix composed of the projection terms ($s \bullet \cos\theta$ and $\sin\theta$), $\boldsymbol{L}$ is the vector of all unknown wall lengths, and **b** is the constant vector composed of the negative values of the known wall projection sums ($-C_H$ and $-C_V$) derived from Equation (7).

To ensure system robustness in cases of conflicting annotations or underdetermined equations, this study introduces a two-tier constraint mechanism. First, the aforementioned room closure principle is applied as a hard constraint. Second, a "pixel reference constraint" is introduced as a soft constraint, requiring that the physical length of unknown walls closely approximate the product of their pixel length and the global scale. This ensures that even isolated regions lacking annotations, such as atria, can obtain reasonable numerical solutions.

Given that measurement errors and geometric simplifications may

Prompt template for room function identification

Role

You specialize in interpreting architectural floor plans and performing functional analysis of spaces.

Task description

Accurately identify the functional category of each individual room based on the provided building floor plan.

Instructions

- Carefully look for and read text or numerical labels inside the room (e.g. "Room 1", "Conference Room", "Office-3"). This is the most direct evidence to determine the function of the room.
- If the room is not labeled with any text, make an inference based on the layout of the furniture, spatial form, and relative location. A large conference table and multiple chairs usually indicates a "meeting room"; a desk and chair combination indicates an "office"; and an open, narrow passageway connecting multiple rooms is a "corridor". An open, long, narrow passageway that connects multiple rooms is a "corridor".
- If a text label conflicts with a visual feature such as a furniture layout (e.g., a room labeled “office" but containing a conference table), the text label is preferred.
- If a space has fuzzy boundaries or too little information to really determine its category, categorize it as “unknown”.
- The output is in json format, key is the room number and value is the category of the determination. {"Room 1": "meeting", "Office-02": "office“}

Fig. 7. Prompt template for extracting room functions.

Prompt template for wall dimension extraction

Role

You specialize in interpreting architectural floor plans and performing functional analysis of spaces.

Task description

The following are two plans of the same building, the first one is a hand-drawn plan with dimensioning, and the second one is a standardized with wall segmentation results, and each wall segment is identified with a wall ID, which is in the format of “W_1”, “W_2”, etc. Extracting wall dimensioning information from original hand-drawn sketches.

Instructions

- The enclosed or semi-enclosed wall topology is used as the primary basis for determining the extent of independent space in a room. A room must be defined by clear wall boundaries. Identify from the hand drawing all walls with dimensions labeled with e.g., 4000mm, 2000mm, etc.
- Ensure that dimensional figures in edge areas are carefully detected; they may be difficult to recognize due to their position against the edge and the vertical alignment of the text. Localized image rotation or enhancement can be done as needed to improve recognition accuracy.Keep all spatial positions and connections exactly the same as the hand-drawn sketch.
- Recognizes walls with explicit dimensioning; if the wall is not dimensioned, it does not need to be recognized.
- Mapping wall IDs to corresponding dimension annotations.
- For each successfully recognized wall segment, output a JSON object containing: { "segment_id": "W_25", "start_point": [x1, y1], "end_point": [x2, y2], "length": 4000}; segment_id: unique identification of the wall; start_point, end_point: pixel coordinates of the wall in the image; length: only output wall segments with explicit numeric labeling according to the assignment of the recognized dimension labeling.

Fig. 8. Prompt template for extracting wall dimensions.

render matrix A underdetermined (fewer equations than unknowns) or ill-conditioned (small perturbations causing significant changes in the solution), Tikhonov regularized least squares is employed to ensure solution stability and physical meaningfulness (e.g., non-negative wall lengths), as shown in Equation (9):

$$min_x \|\mathbf{A} \bullet \boldsymbol{L} - \mathrm{b}\|_2^2 + \lambda \|\boldsymbol{L}\|_2^2 \tag{9}$$

where $\lambda > 0$ is the regularization parameter, which balances the trade-off between the solution's fitting accuracy and its numerical stabilities. In this study, λ is set to 0.0001. Furthermore, to satisfy physical plausibility, the system applies a non-negative constraint to the solution and rounds the results to two decimal places, ensuring that all inferred wall lengths are positive values and comply with standard engineering

conventions.

After solving for the precise physical lengths of all walls, a reference point in the sketch is designated as the origin (0,0) of the physical coordinate system. For single-story buildings, the algorithm defaults to anchoring the bottom-left pixel of the sketch as the physical origin. For multi-story buildings, to ensure inter-story topological consistency, the system requires the user to designate a common vertical reference point (e.g., the corner of an elevator shaft or a standard column position) in the sketches of each floor to serve as the unified global physical origin. A Breadth-First Search (BFS) algorithm is then used to recursively calculate the physical coordinates of each vertex based on the determined wall lengths and orientations, thereby completing the precise mapping from pixel space to physical space.

4) 3D building geometric model generation

Following the construction of the 2D geometry and topology, this step extrudes the model from a 2D plane into 3D space and generates fenestration. First, the method utilizes the floor height information parsed in Task 1 to extrude the 2D polygonal footprint of each room along the Z-axis, thereby creating 3D surfaces such as floors, ceilings, and walls. For exterior walls, the method parametrically generates windows based on the WWR provided in Task 1. The window dimensions are calculated using Equation (10), which assumes that the window maintains the same aspect ratio as its host wall.

$$W_{win} = \sqrt{WWR} \bullet L_{wall}, \quad H_{win} = \sqrt{WWR} \bullet \mathrm{H} \tag{10}$$

Windows are placed by default in the center of their respective walls. Finally, to ensure proper parsing by simulation engines, the vertex order of all surfaces is uniformly adjusted to align their normal vectors outward from the room. This process produces a structured model with precise three-dimensional spatial definitions and window information, with surface definitions directly mappable to the BuildingSurface: Detailed' and 'FenestrationSurface:Detailed' objects in EnergyPlus.

2.3.3. Thermal parameter generator (task 3)

Task 3 aims to identify reference cases from the professional EnergyPlus model database that most closely match the target building's physical characteristics and environmental conditions. Once similar cases are found, their BEM parameters are used to guide the LLM in generating thermal parameters that meet the user's requirements.

1) Similar model retrieval

First, the system performs an initial filtering of the model database based on the key features parsed in Task 1. The similarity between models is quantified using a weighted Euclidean distance, where the weights are set according to the sensitivity of each feature's impact on building energy consumption. To establish a consistent geographical baseline, the system uses the OpenCage API to convert the building's location into coordinates and then maps it to one of the 19 standard climate zones defined by ASHRAE Standard 169. Before calculating the distance, all numerical features undergo logarithmic scaling (Equation (11)) and [0,1] normalization (Equation (12)) to eliminate the influence of differing scales and units.

$$x_{log} = \log(1 + x) \tag{11}$$

$$x_{norm} = \frac{x - x_{log.min}}{x_{log.max} - x_{log.min}} \tag{12}$$

where x is the original feature value, x_{log} is the value after the logarithmic transformation, $x_{log.min}$ and $x_{log.min}$ are the minimum and maximum values of the transformed feature, and x_{norm} is the normalized value, which ranges from [0,1]. Finally, the composite distance is calculated using the weighted Euclidean distance (Equation (13)), which is then converted into a similarity score in the range of [0,1] (Equation (14)). A higher score indicates greater similarity.

$$D_{AB} = \sqrt{\sum_{i=1}^{n} w_i \bullet \left(x_{norm,i} - y_{norm,i}\right)^2} \tag{13}$$

$$S = \frac{1}{1 + D_{AB}} \tag{14}$$

where n is the number of features; w_i is the weight of the i-th feature; $x_{norm,i}$ is the normalized value of the i-th feature for a building model A in the library; $y_{norm,i}$ is the normalized value of the i-th feature for the target building B; D_{AB} is the weighted Euclidean distance, and S is the similarity score, ranging from 0 to 1.

2) Thermal parameter generation

Following the retrieval of the most similar case, detailed information on envelope structures (e.g., wall configurations and material properties) is extracted and dynamically incorporated as contextual knowledge (Context) into a structured prompt. This prompt guides the LLM to generate thermal parameters tailored to the current modeling requirements. Fig. 9 illustrates the prompt construction process for generating exterior wall thermal parameters (designed to meet target WALLU and WALLSP values) based on the wall construction (Construction) and material information (Material) from a similar model. The prompt includes: Objective and Instruction (defining the LLM's role and target parameter metrics), Relevant Model Information (providing the retrieved case's wall layering structure and material properties), Key Points (highlighting physical rules or design preferences to be followed during generation), Calculation Methods (offering formulas for thermal resistance and heat transfer coefficients to ensure accurate reasoning), and Output Format Requirement (specifying JSON). The same prompt construction method is applied to generate thermal parameters for other envelope components, such as roofs, windows, and floors. To ensure the accuracy of the generated parameters, the method incorporates a validation-iteration mechanism. The construction parameters (e.g., thickness of each material layer) generated by the LLM are recalculated to determine the overall heat transfer coefficient. If the deviation from the user-defined target exceeds a preset threshold (defaulting to 0.005 W/(m^2·K)), the system automatically adjusts the prompt (e.g., by imposing stricter constraints on insulation thermal resistance) and initiates a new generation cycle until the precision requirement is met. And to ensure the physical realism and reliability of the generated building constructions, this study establishes "Thermal Parameter Guardrails" for the LLM generation process. When the LLM adjusts construction layers based on the context retrieved via RAG to match the target U-value, it must strictly adhere to predefined material property boundaries. Specifically, the valid ranges are defined as follows: layer thickness 0.01–0.5 m, thermal conductivity 0.01–3.0 W/(m·K), density 50–3000 kg/m^3, and specific heat capacity 100–3000 J/(kg·K). These limits are derived from ASHRAE standards and the statistical distributions of DOE prototype models, preventing the generation of unrealistic constructions (e.g., excessively thin layers or materials with extreme thermal conductivity). If the iteration fails to converge within these boundaries, the system prompts the user to intervene manually or to relax the target requirements.

2.3.4. Operational Parameter Generator (task 4)

Unlike the numerical matching in Task 3, operational parameters (e.g., "working hours 9:00–17:00 with 95% occupancy") require semantic understanding. Therefore, Task 4 employs a RAG method based on semantic similarity. The key advantage of this approach is its ability to go beyond keyword matching to accurately capture the deep semantic relationships between different textual expressions (e.g., "working hours 9:00–17:00" is semantically highly correlated with "regular office hours"). The process begins by using Sentence-BERT [67] to generate semantic vector representations of both the user's textual description and the various operational profiles stored in the model database, converting the text into high-dimensional vectors. Subsequently, it retrieves

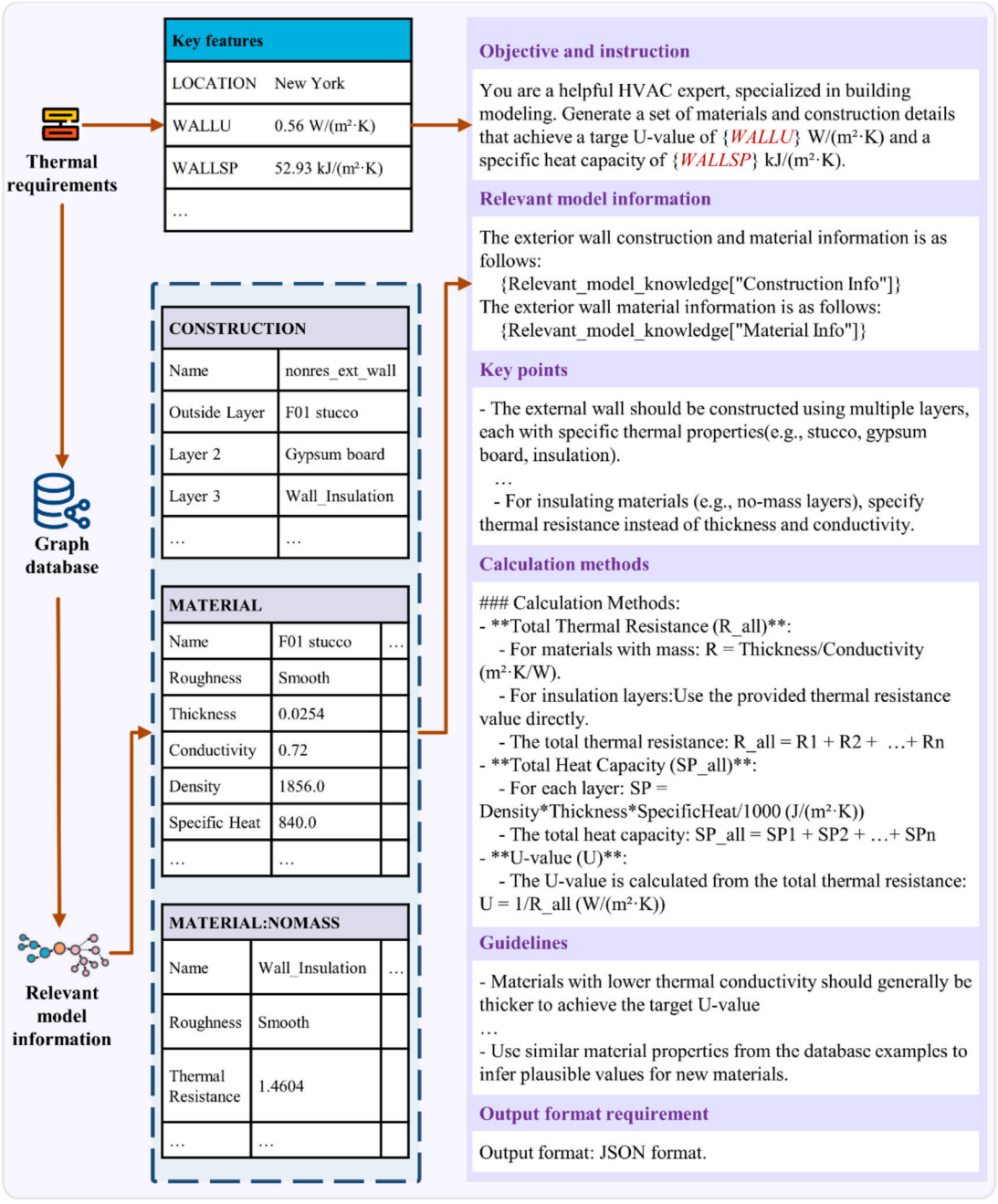


Fig. 9. Prompt template for thermal parameters generation (external wall).

the most semantically similar reference schedule by calculating the cosine similarity (Equation (15)) between the user's input vector v_1 and the vectors v_2 in the library.

$$cosine_similarity(v_1, v_2) = \frac{v_1 \bullet v_2}{\|v_1\| \|v_2\|} \tag{15}$$

To ensure the applicability of the retrieval results, the process pre-filters the database based on the building type (e.g., "Office Building," "Residential Building") parsed in Task 1. This step prevents mismatches in operational patterns that could arise from differing building functions. After retrieving the most similar reference schedule, this information is injected as contextual knowledge into a structured prompt (as shown in Fig. 10). This guides the LLM to generate a complete, full-period schedule in a standardized JSON format, based on both the user's input and the retrieved case. Finally, a post-processing module optimizes the generated schedule by merging adjacent time segments with identical occupancy rates, producing a more concise timetable.

2.3.5. IDF generator and simulator (task 5)

A specialized script maps the structured outputs from the previous tasks, including geometry (Task 2), thermal properties (Task 3), and schedules (Task 4), to specific EnergyPlus Input Data File (IDF). Specifically, this script accurately maps this structured data to specific Objects and Fields within the IDF file. For instance, geometric vertex data is converted into 'BuildingSurface:Detailed' objects, wall construction and material information are mapped to 'Construction' and 'Material' objects, and operational schedules are formatted as 'Schedule: Compact' objects. The script then automatically triggers the EnergyPlus application programming interface (API) to execute the simulation and output hourly load curves.

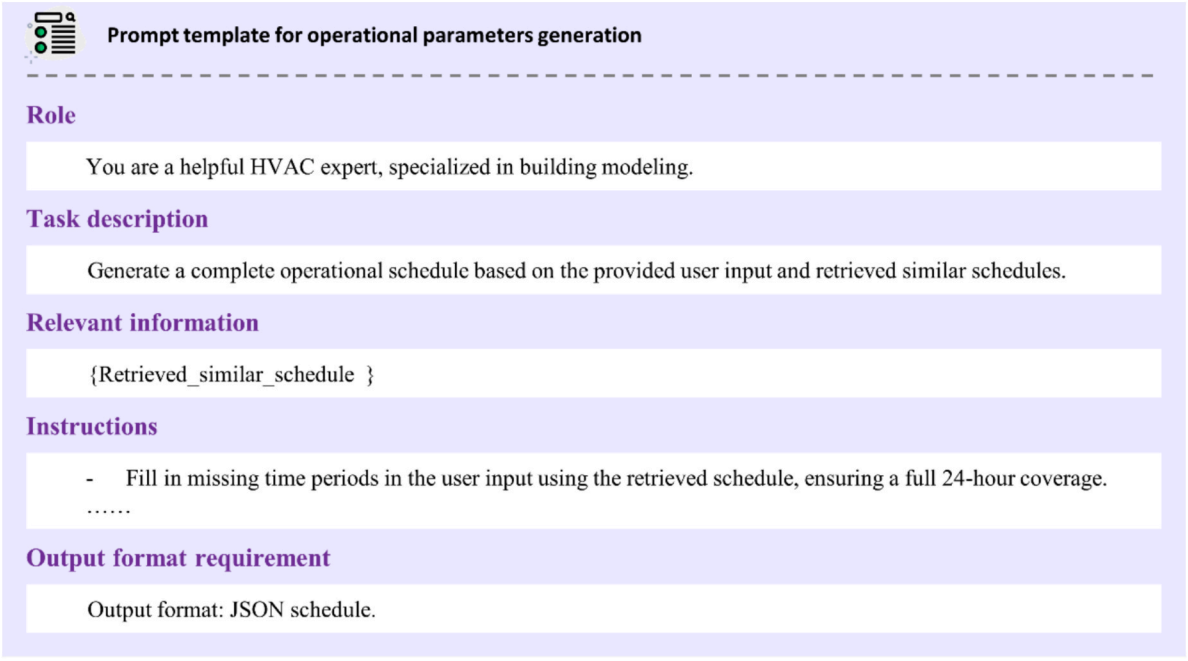


Fig. 10. Prompt template for operational parameters generation.

3. Validation and analysis

This section aims to provide a comprehensive and systematic evaluation of the proposed method's performance. First, Section 3.1 details the construction and statistical characteristics of the EnergyPlus model database used for the retrieval tasks. Section 3.2 introduces the case study for validation. Subsequently, Section 3.3 validates the robustness of the in-context learning (ICL)-based Requirements Parser, and Section 3.4 assesses the accuracy of the Building Geometry Generator. Sections 3.5 and 3.6 then examine the accuracy of the Thermal and Operational Parameter Generators, respectively. Finally, Section 3.7 provides a direct comparison of method's overall performance against a benchmark model, comprehensively evaluating its practical value from the perspectives of simulation accuracy and modeling efficiency.

The experiments in this study adopted a hybrid model deployment strategy to match the technical requirements of different tasks. For the purely text-based analysis and generation tasks (Task 1, 3, and 4), Meta's open-source Llama-3-8B-Instruct model was used. This model was deployed on a local workstation equipped with an NVIDIA GeForce RTX 3090 GPU (24 GB GDDR6X VRAM) to support the method's inference and generation processes. For the core Building Geometry Generator (Task 2), which involves the visual understanding and reconstruction of sketches, industry-leading multimodal LLMs were utilized, specifically OpenAI's GPT-4o and Google's Gemini-2.5-Pro.

3.1. EnergyPlus validation model database

The RAG module of this study relies on a high-quality, diverse knowledge base. To this end, an EnergyPlus model database was constructed based on the U.S. Department of Energy (DOE) Commercial Prototype Buildings dataset [18]. This dataset is well-suited as a reference for the proposed method for the following reasons: 1) **Standardization and quality**. All models are built according to ASHRAE standards [68] and meticulously calibrated, ensuring parameter reliability and consistency [69]. 2) **Diversity and representativeness**. The dataset systematically encompasses 19 ASHRAE standard climate zones and various typical commercial building types [70], representing a wide range of energy consumption scenarios. 3) **Data compatibility**. The models contain rich, parseable key features and detailed construction information, fully meeting the method's requirements for similarity retrieval and context generation [71]. Ultimately, this study parsed 228 standard models, storing them in a structured format within a Neo4j (v5.20.0) graph database [72].

The graph schema of the database is centered on the "BuildingModel" node, which connects to the "ClimateZone" node via the "HAS_CLIMATE" relationship. Furthermore, it associates key physical features (e.g., AREA, NL, CR) and detailed thermal and operational attributes (corresponding to the para and para_detail sub-nodes) through the "HAS_PARAMETER" and "HAS_DETAIL_PARAMETER" relationships, respectively. This storage strategy anchors the target model using key geometric features and subsequently employs graph traversal to accurately extract the building's construction details and operational schedules.

During similarity retrieval, the dynamic calculation of feature weights is implemented via the Cypher query language. A typical retrieval query logic is as follows: "*MATCH (m:BuildingModel)-[:HAS_CLIMATE] −> (z:ClimateZone {ZONE_ID: '3A'}) RETURN m ORDER BY apoc.coll.sum ([m.AREA * 0.04, m.NL * 0.02, …, m.CR * 0.1]) DESC LIMIT 5.*" By calculating the weighted Euclidean distance between the target building and the sample models in the library, this mechanism identifies the top five most similar reference models, thereby providing reliable physical constraint contexts for Tasks 3 and 4.

To quantify the internal characteristics of the model database, Table 2 provides a detailed statistical description of the extracted key features, and Fig. 11 presents their frequency distribution. Analysis reveals that the database covers a range of building sizes from small (464 m^2) to large (approximately 20,000 m^2), with the majority concentrated in the medium range (mean 5,178 m^2). The low average CR (mean 0.257) and moderately low WWR (predominantly ranging from 0.14 to 0.27) indicate that the library's models generally adopt energy-efficient, compact geometric designs. Thermal performance of envelope structures (such as an average WALLU of 1.078 W/(m^2·K) and ROOFU values mostly below 0.2 W/(m^2·K)) suggests excellent insulation performance across most buildings, making them suitable for energy efficiency studies. Overall, the parameter distribution of this model database not only spans diverse design scenarios but also aligns closely with the principles of energy-efficient building design. Its authoritative, diverse, and high-quality attributes provide a robust and reliable data foundation

Table 2
Overall distribution of key features.

	AREA	NL	CR	WWR	WALLU	WALLSP	ROOFU	WINDOWU	SLABU
count	228	228	228	228	228	228	228	228	228
mean	5177.85	4.167	0.257	0.208	1.078	50.500	1.331	1.985	0.235
std	5671.57	3.395	0.090	0.094	1.096	29.219	1.953	0.540	0.325
min	464.690	1	0.083	0.066	0.281	0.000	0.163	1.363	0.000
25%	1837.52	2	0.220	0.137	0.512	50.497	0.186	1.408	0.062
50%	3077.48	3	0.262	0.198	0.693	67.329	0.228	1.942	0.156
75%	5195.02	4.5	0.279	0.267	1.076	67.329	1.389	2.408	0.224
max	19890.10	14	0.458	0.400	5.430	67.329	4.706	3.009	1.304

Note: Std represents standard deviation; 25%, 50%, and 75% denote the 25th percentile, median, and 75th percentile, respectively.

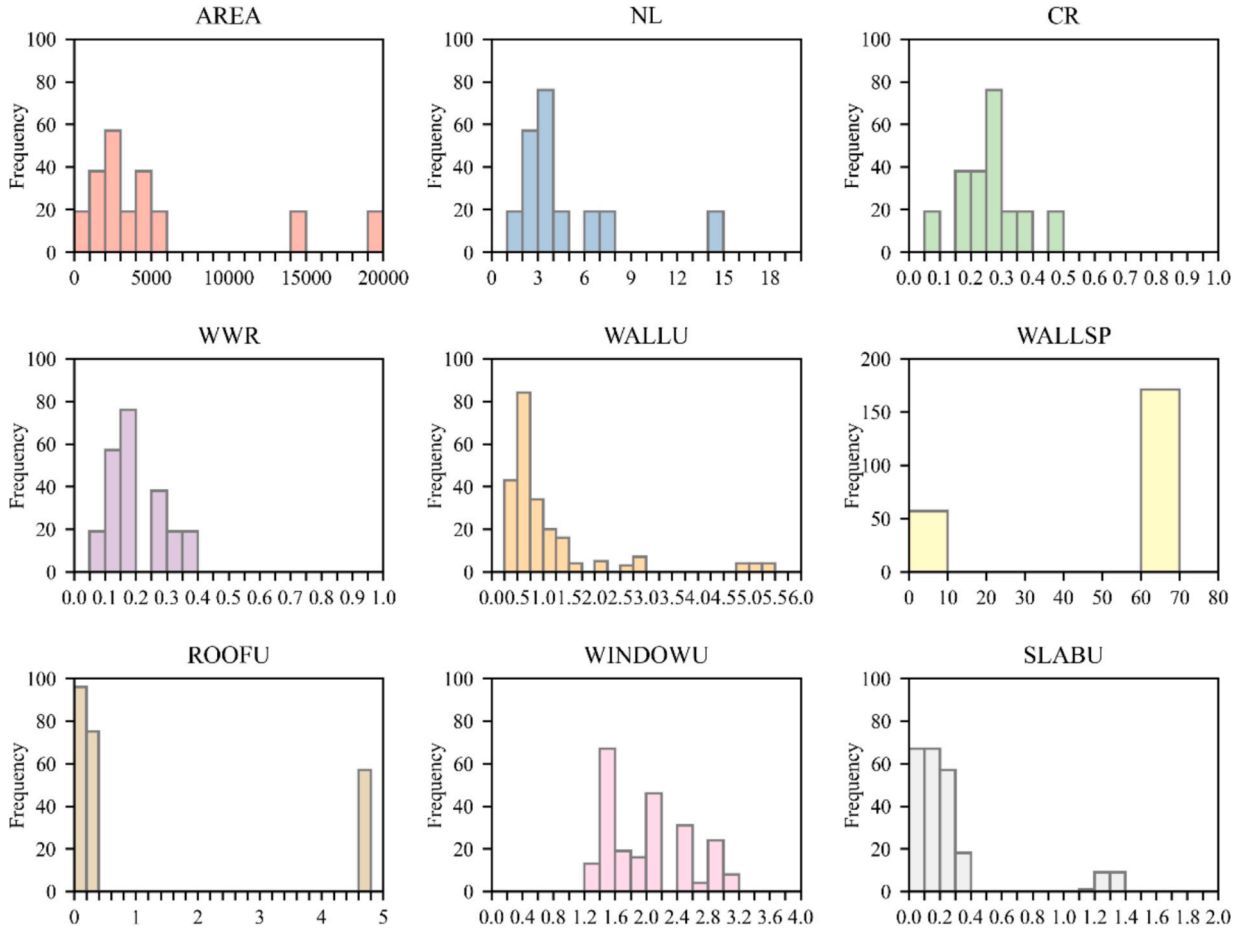


Fig. 11. Distribution histogram of key features in the EnergyPlus model database.

for the subsequent RAG tasks of the automated BEM method.

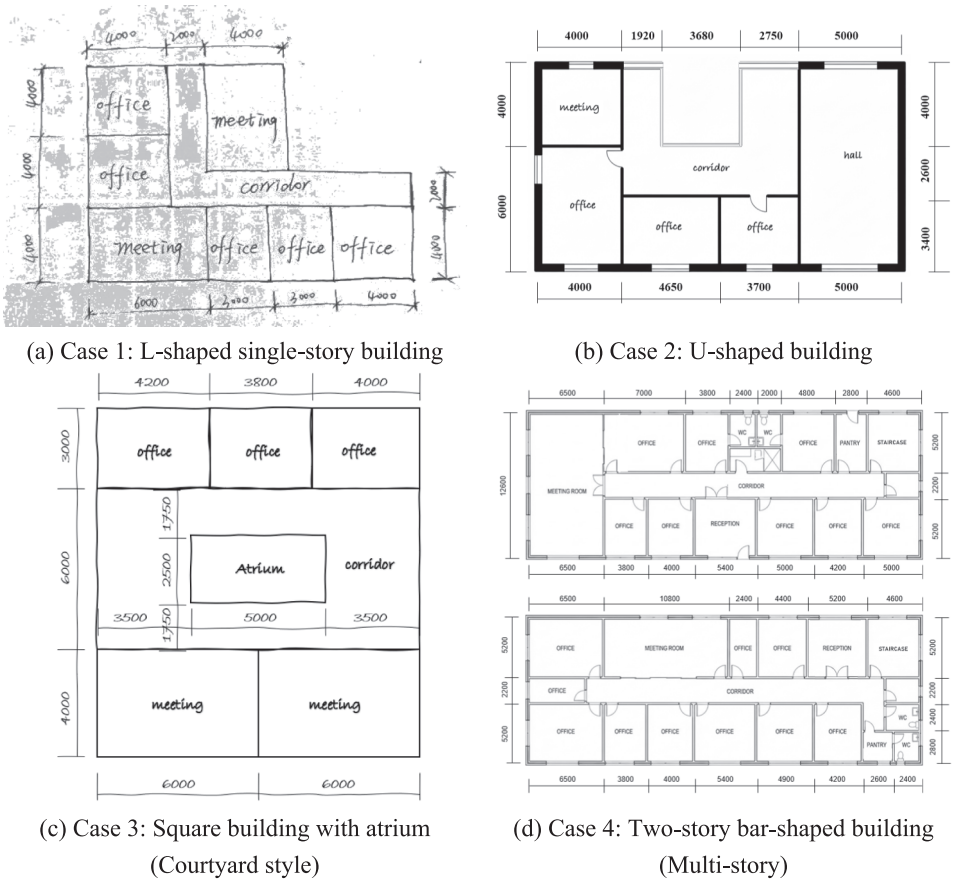
3.2. Case introduction and workflow input

To systematically evaluate the generalization ability and robustness of the proposed automated BEM method across diverse building morphologies and climatic conditions, this study selected four building cases with distinct morphological differences for validation. These buildings are located in different typical ASHRAE climate zones and encompass non-convex plans (L-shaped and U-shaped), courtyard-style layouts (square with atrium), and multi-story structures (two-story bar-shaped), thereby comprehensively covering common spatial organization types encountered in the early design stage.

The key parameters of each case are summarized in Table 3. The input for all cases consists of architectural floor plan sketches in image format (as shown in Fig. 12) and corresponding modeling requirements

Table 3
Key design parameters of the four typical validation cases.

ID	Case 1	Case 2	Case 3	Case 4
Morphology	L-shape	U-shape	Courtyard	Bar-shape
Location	Shanghai	Chongqing	Beijing	Guangzhou
Climate Zone	3A	3A	4A	2A
H	3	3.5	3	3.2
AREA	150	158.88	156	853.63
NL	1	1	1	2
CR	0.69	0.68	0.65	0.37
WWR	0.29	0.294	0.34	0.297
WALLU	0.75	0.97	0.46	0.73
WALLSP	57.6	42.42	224.14	85.92
ROOFU	0.18	0.19	0.37	0.88
WINDOWU	2.02	2.26	1.2	3.11
SLABU	1.8	3.84	3.75	3.62



(a) Case 1: L-shaped single-story building

(b) Case 2: U-shaped building

(c) Case 3: Square building with atrium (Courtyard style)

(d) Case 4: Two-story bar-shaped building (Multi-story)

Fig. 12. Input architectural floor plan sketches for the four typical validation cases with diverse morphologies.

expressed in natural language. The floor plan sketches define the building's basic boundaries, room topology, and room functions, while the modeling requirements specify the simulation performance targets typical of the early design stage. No BIM or CAD geometric data is introduced throughout the process. As an example of the input format, the complete natural language modeling requirements for Case 1 (L-shaped single-story building) are presented in Table 4. The remaining three cases employ natural language descriptions with the same structure, including consistent thermal performance targets, functional zoning requirements, and operational schedules.

3.3. Requirements Parser robustness validation

This section quantitatively evaluated the robustness and accuracy of the Requirements Parser (Task 1) in handling diverse natural language inputs, as its performance underpins the successful initiation of the entire automated BEM method. Due to the semantic and syntactic variability in user inputs (e.g., "window-to-wall ratio of 40%" versus "proportion of windows to walls set to 0.4"), traditional rule-based or regular expression methods struggle to address these challenges effectively. In contrast, LLM employed in this study leverages its robust semantic understanding to overcome this limitation.

A test set comprising 360 samples was constructed to simulate real-

Table 4
Example of textual modeling requirements for Case 1 (L-shaped single-story office building).

This office building is located in the vibrant city of Atlanta, covering an area of approximately 150 square meters and a ceiling height of 3 m. The design of the building adopts an efficient compactness ratio of 0.69. Its window to wall ratio is 0.29, and the heat transfer coefficient of the windows is 2.02 W/(m^2·K), providing sufficient natural light while minimizing energy loss. The heat transfer coefficient of the external wall is 0.75 W/(m^2·K), the specific heat capacity is 57.6 kJ/(m^2·K), the heat transfer coefficient of the floor is 1.8 W/(m^2·K), and the heat transfer coefficient of the roof is 0.18 W/(m^2·K), which meets energy-saving standards. This one-story building exemplifies a sophisticated and energy-conscious architecture, tailored to meet modern sustainability standards while providing a comfortable working environment for its occupants.

Room types and occupancy schedules:

1) Office:
- Occupancy rate: 10 m^2 per person; lighting power density: 12 W/m^2, equipment power density: 11 W/m^2.
- Air Conditioning Settings: 24°C(summer); 22°C(winter)
- Schedule: the occupancy rate increases to 10% at 09:00; peak at 95% from 10:00 to 11:00, reduces to 80% by 12:00; returns to 95% from 13:00 to 17:00, drops to 30% by 18:00.

2) Meeting:
- Occupancy rate: 2.5 m^2 per person; lighting power density: 15 W/m^2; equipment power density: 15 W/m^2.
- Air Conditioning Settings: 22°C(summer); 20°C(winter)
- Schedule: 0% occupancy before 09:00; 50% by 10:00, peaks at 95% by 11:00; drops to 30% from 11:00 to 12:00, 10% by 13:00; 50% by 14:00, increases to 60% by 16:00; reduces to 20% by 18:00, 0% after 18:00.

3) Corridor:
- Occupancy rate: 8 m^2 per person; lighting power density: 5 W/m^2.
- Air Conditioning Settings: 24°C(summer); 18°C(winter)
- Schedule: 0% occupancy before 08:00; 50% by 09:00, drops to 20% by 11:00; peaks at 80% by 13:00; reduces to 40% by 17:00, 10% by 18:00; 0% after 24:00.

world application scenarios. The samples are categorized into two types: 1) single-feature samples, consisting of 30 distinct statements for each of the 11 key features, designed to reflect varied expressions; and 2) composite-feature samples, comprising 30 complex statements that simultaneously include all 11 features, thereby testing the module's comprehensive parsing ability under information-dense inputs. The evaluation metric adopted is accuracy (ACC), defined as per Equation (16):

$$\mathrm{ACC} = N_T/N \tag{16}$$

where N_T is the number of correctly parsed samples, and N is the total number of samples.

Experimental results (as shown in Fig. 13) indicate that the module exhibited only 2 parsing errors across 360 tests, achieving an accuracy of 99.4%. This outcome confirms the robustness of the Requirements Parser based on Llama-3-8B-Instruct, demonstrating its ability to reliably transform diverse unstructured user requirements into standardized inputs essential for subsequent similarity model retrieval and parameter generation tasks.

3.4. Building geometry generator effectiveness evaluation

To comprehensively evaluate the performance of the Building Geometry Generator (Task 2), this section first presents a qualitative comparison to demonstrate the performance advantages of the proposed multi-technology fusion workflow over single-technology approaches (i.e., LLM-only and DeepFloorplan-only) in geometric reconstruction tasks. Subsequently, Section 3.4.2 conducts a quantitative evaluation of geometric extraction accuracy across four distinct building cases (detailed in Section 3.2) to further verify the robustness and applicability of the method. Section 3.4.3 then analyzes the sensitivity of the regularization parameter λ to determine the optimal configuration for coordinate inference. Finally, Section 3.4.4 documents typical failure modes encountered in extreme scenarios and provides practical user guidance for drawing optimization.

3.4.1. Qualitative comparison of geometry reconstruction approaches

This section uses the typical L-shaped office building case (Case 1) to visually demonstrate the multi-stage processing workflow of the proposed automated BEM method. Furthermore, a series of comparative experiments were designed to systematically benchmark the proposed method against current mainstream single-technology pathways. The comparative approaches include:

1) **Automated BEM method**: The complete multi-stage workflow proposed in this study, which integrates multimodal LLMs, computer vision techniques, and computational geometry algorithms. 2) **LLM-only**: Serving as a baseline method, this approach relies solely on a single multimodal LLM (GPT-4o) to directly parse hand-drawn sketches into vector graphics and output SVG code, aiming to assess the geometric spatial reasoning and reconstruction capabilities of current multimodal LLMs. 3) **DeepFloorplan-only**: As another baseline method, this approach depends exclusively on the computer vision model (DeepFloorplan) to highlight the inherent limitations of standalone computer vision techniques in handling hand-drawn sketches with noises.

The processing flow and intermediate results of the proposed method are illustrated in Fig. 14, using a representative L-shaped building as an example. First, GPT-4o denoises errors in the original hand-drawn sketch, effectively filtering out line noise and generating a topologically clear SFP (Fig. 14(a)). Although the output at this stage preserves the original spatial layout in terms of topology, it still exhibits issues of local proportional distortion (e.g., the visual widths of the vertical and horizontal corridors are inconsistent); this issue is corrected in subsequent steps. Next, the DeepFloorplan model extracts the pixel masks for all walls (Fig. 14(b)). Based on these masks, skeletonization and the Probabilistic Hough Transform are applied to initially detect 41 vector line segments (Fig. 14(c)). After optimization through collinear merging and endpoint extension, a final set of 27 wall segments is obtained (Fig. 14(d)). Based on the graph structure constructed from these vectorized walls, a Depth-First Search (DFS) algorithm successfully identified all 8 closed room units without any duplicates or omissions (Fig. 14 (d)). In the dimension inference and coordinate mapping stage, Gemini-2.5-Pro was utilized to successfully extract 12 explicit dimensional annotations from the original sketch. These known dimensions, combined with the geometric closure constraints of the rooms, formed a sparse system of linear equations. By solving this equations system, the method accurately inferred the physical lengths of the remaining 15 unknown wall segments. This step not only completed the geometric information but also systematically corrected the proportional distortions introduced during the SFP stage, ultimately generating a dimensionally-calibrated 2D geometric model that is fully consistent with the design intent (Fig. 14(e)). Finally, by incorporating the layer height and WWR parsed in Task 1, the method completed the 3D model extrusion and parametric window generation, producing a simulation-ready building geometric model (Fig. 14(f)).

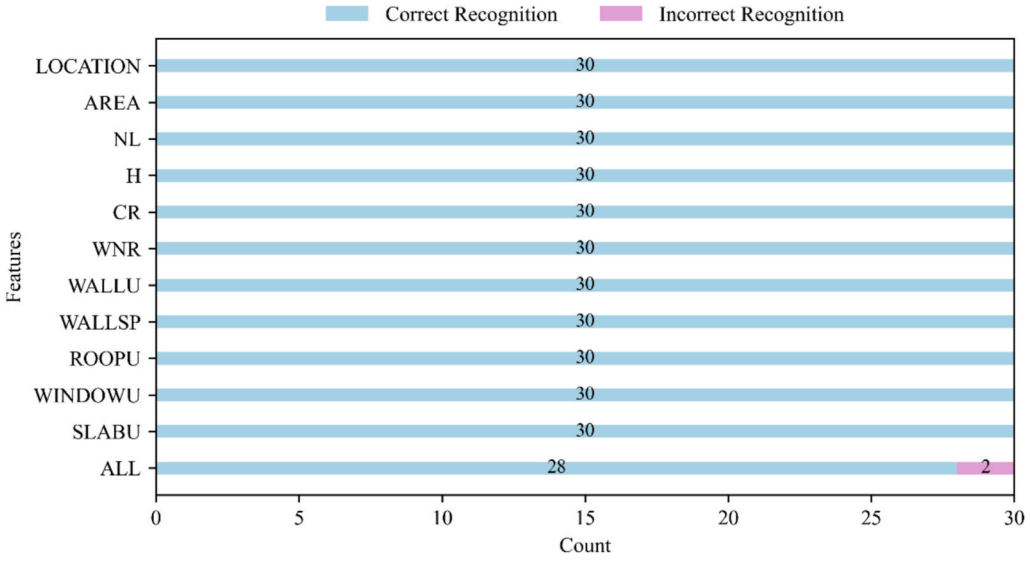
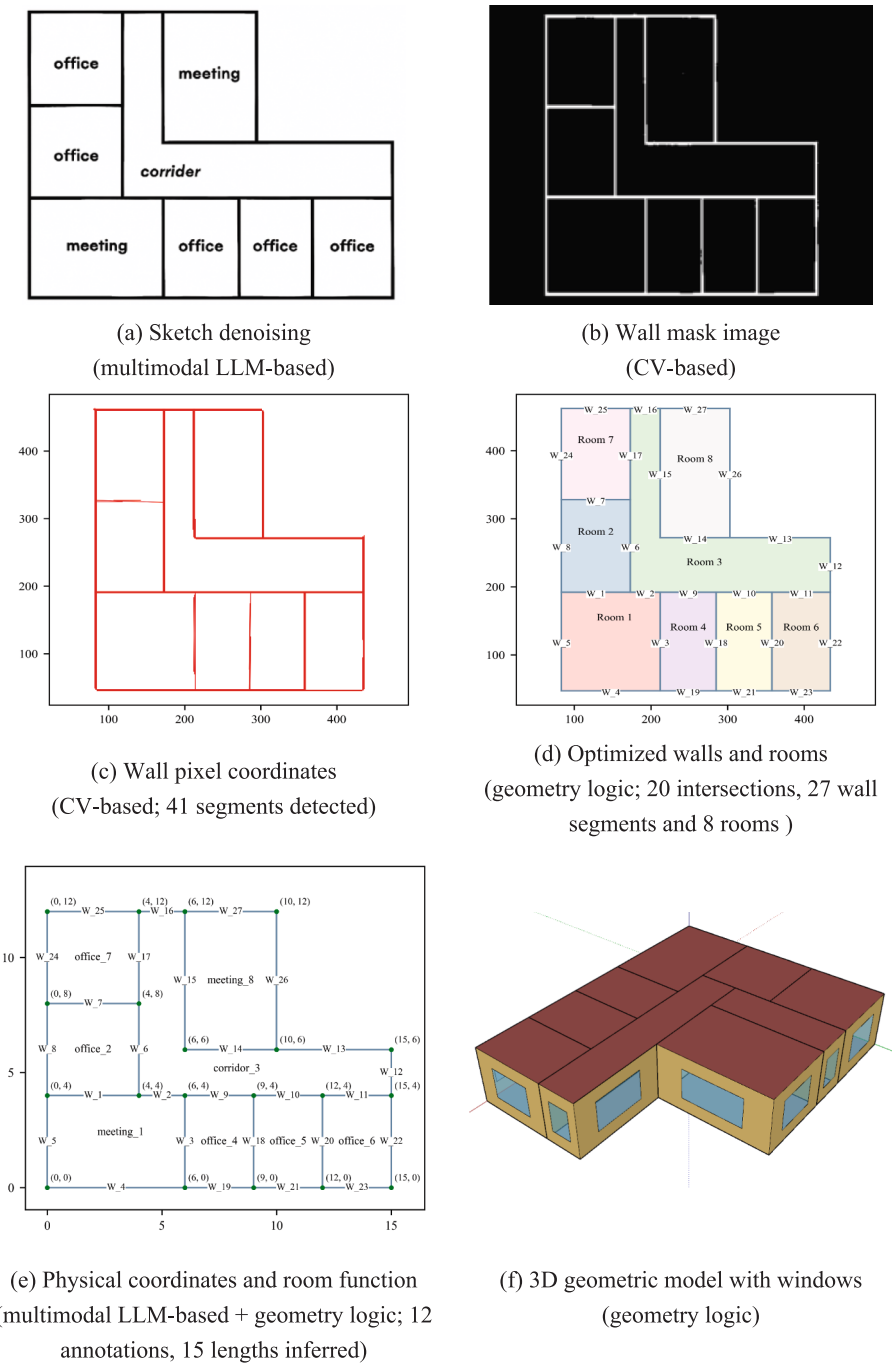


Fig. 13. Robustness test results of Requirements Parser.



(a) Sketch denoising (multimodal LLM-based)

(b) Wall mask image (CV-based)

(c) Wall pixel coordinates (CV-based; 41 segments detected)

(d) Optimized walls and rooms (geometry logic; 20 intersections, 27 wall segments and 8 rooms)

(e) Physical coordinates and room function (multimodal LLM-based + geometry logic; 12 annotations, 15 lengths inferred)

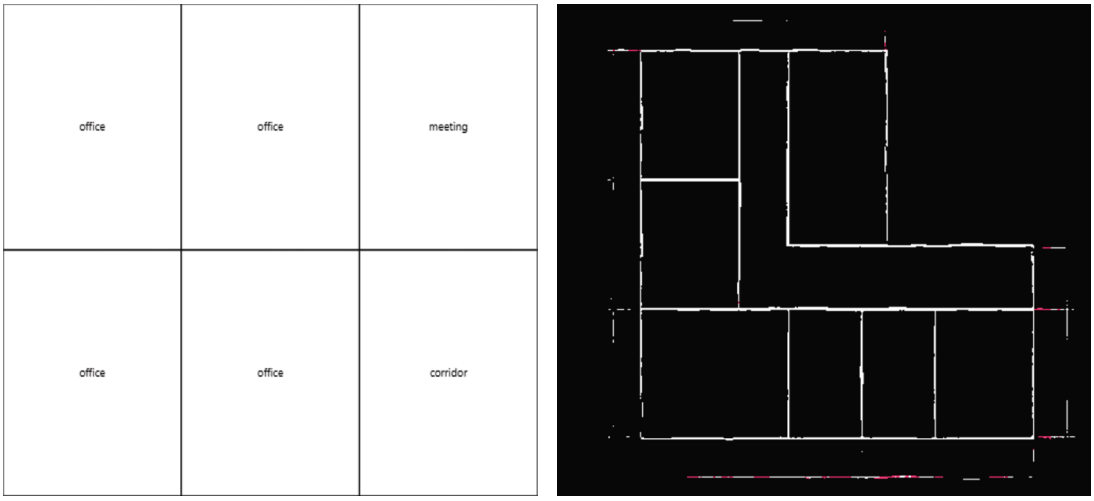
(f) 3D geometric model with windows (geometry logic)

Fig. 14. The multi-stage processing workflow of the Building Geometry Generator within the proposed automated BEM method.

As shown in Fig. 15, both baseline methods failed to successfully complete the geometric reconstruction task. The LLM-only output (Fig. 15(a)) produced severe geometric deviations. Although LLM-only approach generated syntactically correct SVG code, the content erroneously simplified the original L-shaped complex building into a regular 2 × 3 grid, completely losing the core architectural topology and geometric features. Similarly, the DeepFloorplan-only approach (Fig. 15(b)) proved sensitive to noise, incorrectly identifying dimension annotation lines in the hand-drawn sketch as wall structures and failing to interpret any semantic information (e.g., room functions or dimension values). Consequently, it was incapable of constructing a simulation model that contained complete physical and functional information.

These comparative experiments clearly demonstrate that single-technology pathways, whether LLM-only or DeepFloorplan-only, exhibit significant shortcomings when processing the complex task of hand-drawn sketches. In contrast, the integrated multi-technology approach adopted by the proposed automated BEM method successfully overcomes the limitations of these single-method approaches. By combining the macro-level understanding of multimodal LLMs, the pixel-level segmentation of deep learning, and the precise reasoning of computational geometry, the method achieves a robust conversion from an unstructured sketch into a structured, high-fidelity 3D geometric model.

To further validate the generalization capability of the algorithm across buildings of varying morphologies and scales, Fig. 16 presents the reconstruction results for Case 2, Case 3, and Case 4. Given that the reconstruction workflow for these cases mirrors that of Case 1, only the 2D coordinate plots (incorporating inferred physical dimensions and



(a) LLM-only baseline: failure in spatial reasoning and topological preservation

(b) CV-only (DeepFloorplan) baseline: failure in noise filtering (interpreting dimensions as walls) and semantic extraction.

Fig. 15. Comparison of the limitations of single-technology approaches in the geometric reconstruction task.

room function labels). The results demonstrate that the proposed method achieves a robust conversion from sketches to 3D models, even when confronting complex challenges such as the multiple concave corners in Case 2, the inner boundary extraction in Case 3, and the cross-floor topological correlations in Case 4. This confirms the method's adaptability to a diverse range of architectural prototypes.

3.4.2. Quantitative evaluation of geometric accuracy

This section aims to evaluate the accuracy and robustness of the proposed method in generating geometric models from hand-drawn sketches through quantitative error analysis, determining whether it meets the precision requirements for energy simulation in the early design stage. To ensure the objectivity of the evaluation, precise "Ground Truth" benchmark models were established for each of the four cases introduced in Section 3.2. These benchmarks were manually constructed in EnergyPlus by experts based on all unambiguous dimensional annotations in the sketches, as well as the explicit layer height and WWR parameters specified in the modeling requirements. Based on this, a systematic comparative error analysis was conducted on the geometric reconstruction results generated by the automated method. The evaluation metrics are defined as follows:

1) Room Area and Perimeter Error (%): The relative deviation of the generated room area/perimeter from the benchmark values.
2) Wall Length Error (m): The mean absolute error between all reconstructed wall segments and their design annotations.
3) WWR Error (%): The percentage error of the generated fenestration area relative to the ground truth.
4) Window Placement Offset (m): The Euclidean distance between the center point of each generated window and its benchmark position.
5) Room Function Classification Accuracy (%): The proportion of rooms with explicit textual labels in the sketch that were correctly identified by the multimodal LLM.

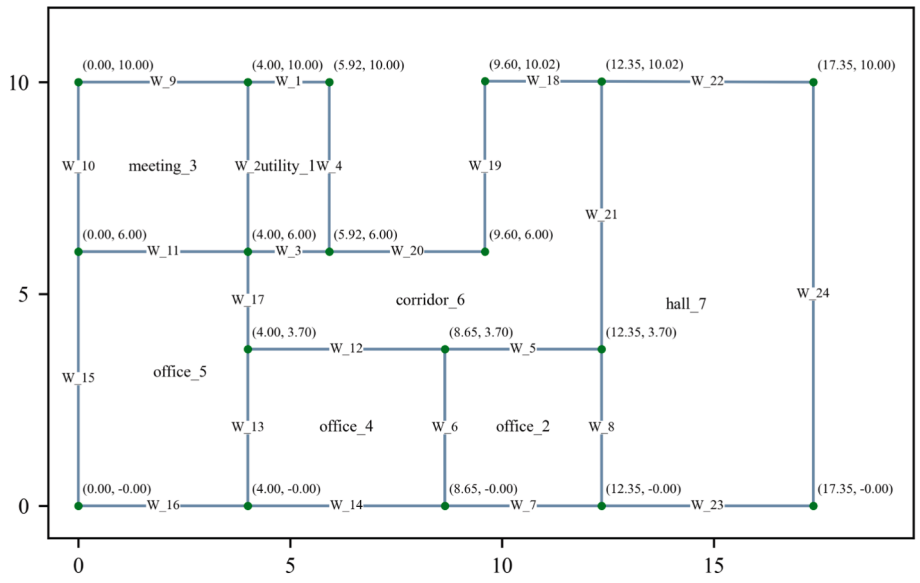
For early-stage design analysis, the tolerance targets set in this study are (reference [73]): Room Area/Perimeter Error $< 10\%$, Wall Length Error < 0.3 m, WWR Error $< 10\%$, Window Offset < 0.3 m, and Room Function Classification Accuracy $> 90\%$.

Table 5 summarizes the statistical results of the geometric metrics for the four cases. The results indicate that the proposed method maintains reliable geometric fidelity across various morphologies (including single-story, multi-story, regular, and irregular layouts). For the simpler morphologies of Case 1 and Case 3, the method achieved precise reconstruction with minimal deviation. While there were certain errors in the more complex U-shaped (Case 2) and multi-story bar-shaped layouts (Case 4), the overall average errors remained far below the set tolerance targets. Notably, the average WWR error was low (0.32%); this is because the algorithm reverse-calculates window dimensions based on the target value, with minor residuals primarily stemming from vertex coordinate rounding and floating-point calculation limitations. The average accuracy of room function classification reached 91.2%. Misclassifications mainly occurred in rooms lacking text labels in the sketches; for instance, an unlabeled corridor in Case 2 was misclassified as a "Utility" room, and an empty room with no usage label in Case 4 was misclassified as an "Office" by the LLM based on the surrounding layout logic. Additionally, the label "Atrium" in Case 3 was written too illegibly and was misidentified as "Atnum." Since misclassifications mostly occurred in auxiliary spaces and the overall accuracy exceeded 90%, the system can still provide reliable support for the subsequent automatic assignment of parameters.

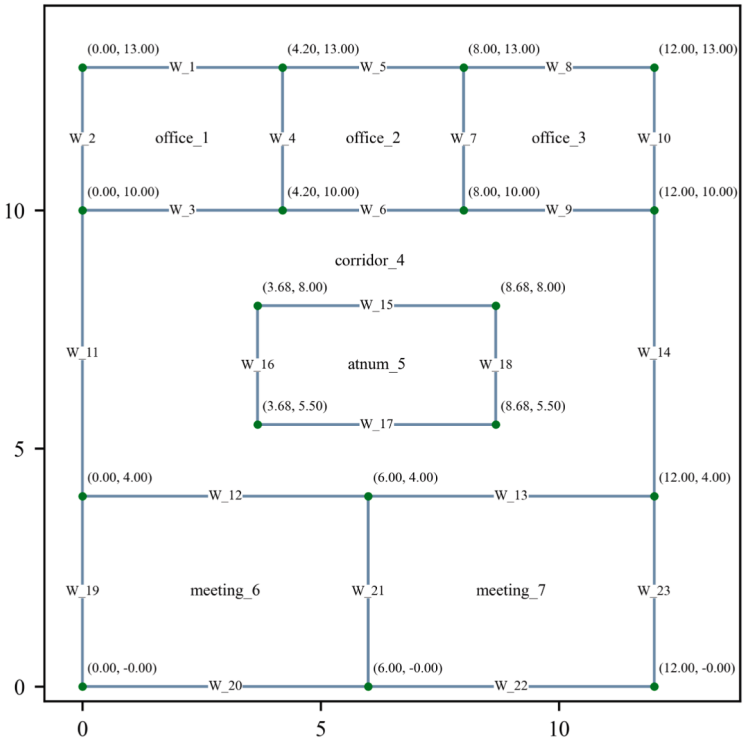
To further reveal error distribution characteristics, Fig. 17 presents boxplots of room area error, perimeter error, wall length error, and window offset for all room samples. The boxplots show that the median errors for all metrics are tightly clustered around the zero line. The narrow Interquartile Range (IQR) for Case 1 and Case 3 reflects the algorithm's robust certainty when processing standard single-story layouts. Note that since the original sketches for Case 1 and Case 3 did not depict windows, window placement comparisons could not be performed, and their window offsets are recorded as null in the statistics.

Combining the boxplot distributions (Fig. 17) with the underlying logic of the algorithm, the causes of deviations in Case 2 and Case 4 can be categorized as follows:

1) **Chain inference deviations caused by missing annotations**: In Case 4 (the multi-story case), internal walls of some auxiliary rooms (e.g., WC, Pantry) and rooms at the ends of corridors lacked explicit annotations. According to the algorithm's logic, the lengths of such walls must be inferred relying on the pixel scale. Non-uniformity in the scale of the original hand-drawn floor plan directly led to wall length errors, which in turn caused fluctuations in room area and



(a) Case 2: U-shaped building handling multiple concave corners

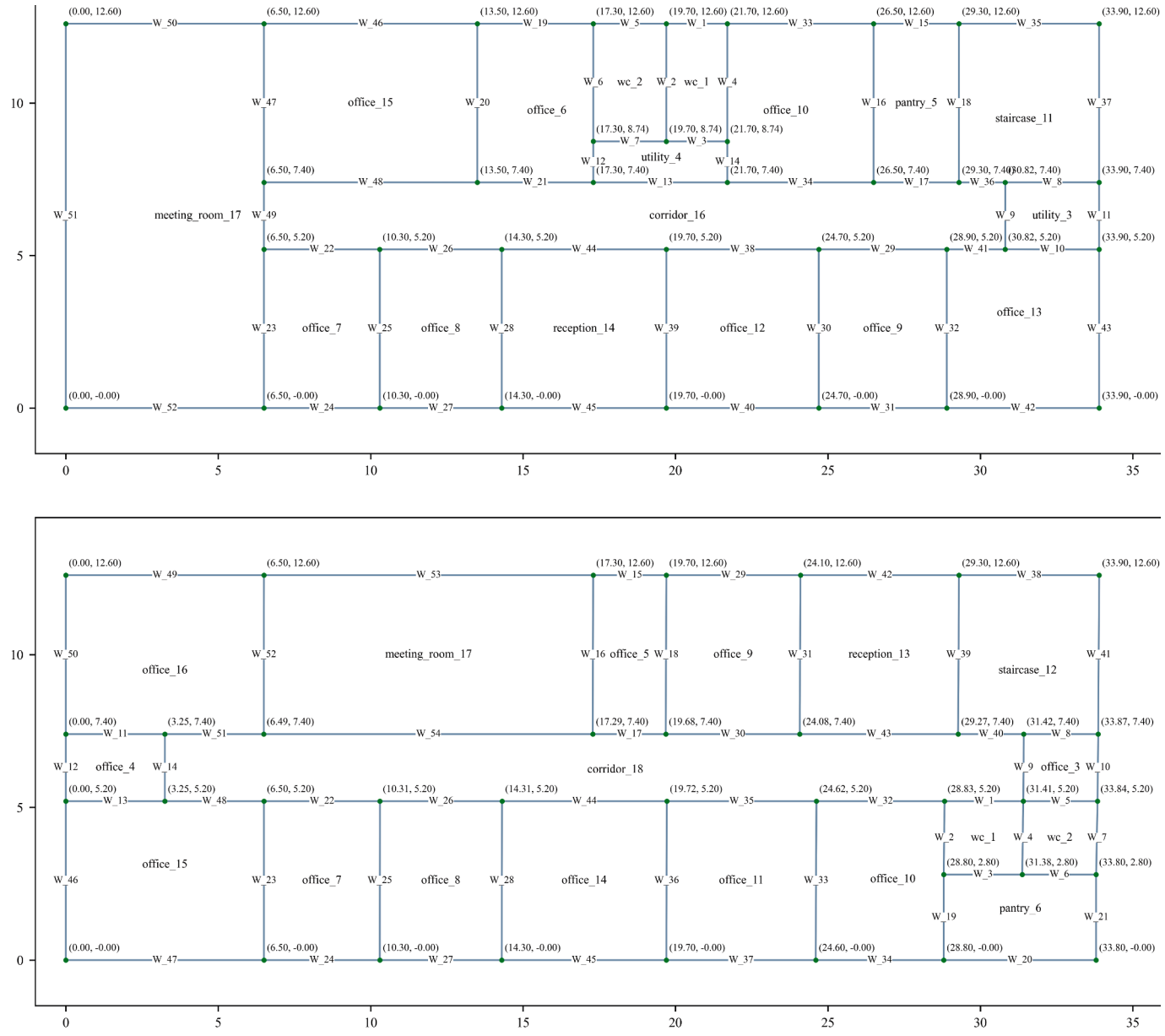


(b) Case 3: Courtyard building handling inner boundary extraction

Fig. 16. Geometric reconstruction results for validation cases with diverse morphologies in 2D physical coordinates.

perimeter through geometric closure constraints. Simultaneously, as windows follow an automatic centering logic, offsets in wall lengths and endpoint coordinates directly resulted in shifts in the window centroids, raising the average offset in Case 4 to 0.098 m.

2) **Topological association failure and dimension misinterpretation**: In both Case 2 (U-shaped) and Case 3 (Courtyard), local accuracy degradation occurred because annotation information could not be effectively mapped to the topological structure. In Case 2, although the corridor width (2400 mm) was clearly labeled on the outside of the sketch, the annotation was not explicitly topologically linked to a specific wall ID. Consequently, the algorithm degenerated to a soft constraint solution based on pixel ratios. Similarly, in Case 3, the dimension annotation for the internal span of the Atrium lacked a binding to specific wall boundaries, preventing it from being incorporated into the hard constraint equation system. This failure in association forced the algorithm to rely on the less robust pixel



(c) Case 4: Multi-story building geometry with cross-floor topological correlations

Fig. 16. (*continued*).

Table 5
Summary of quantitative geometric accuracy metrics across four cases (mean ± standard deviation).

Metric	Case 1	Case 2	Case 3	Case 4	Mean	Tolerance target
Mean Room Area Error (%)	0.01 ± 0.01	1.15 ± 1.4	0 ± 5	7.30 ± 0.3.80	2.97	< 10%
Mean Room Perimeter Error (%)	0.003 ± 0.003	0.44 ± 4353	0 ± 3	3.89 ± 9349	1.08	< 5%
Wall Length Error (m)	0.0002 ± 0.00003	0.028 ± 0.0041	0 ± 4	0.189 ± 0.1429	0.05	< 0.3 m
Mean Window Offset (m)	–	0.004 ± 0.006	–	0.098 ± 0.264	0.03	< 0.3 m
WWR Error (%)	0.34	0.17	0.44	0.34	0.32	< 10%
Room Function Classification Accuracy (%)	100 (8/8)	85.71 (6/7)	85.71 (6/7)	97.14 (34/35)	91.2	> 90%

reference, which was the primary cause of coordinate deviations in corridors, atria, and adjacent rooms in these cases.

3) **Relative error amplification effect in small-scale spaces**: The maximum area error in Case 4 reached 44.01%. Tracing back, it was found that such extreme values appeared in small spaces with an area of less than 8 m^2. Due to the small base value, even a minute physical displacement (approx. 0.2 m) caused by missing annotations resulted in a significant relative percentage error.

Despite these local outliers, 75% of the sample points (Q3) fell

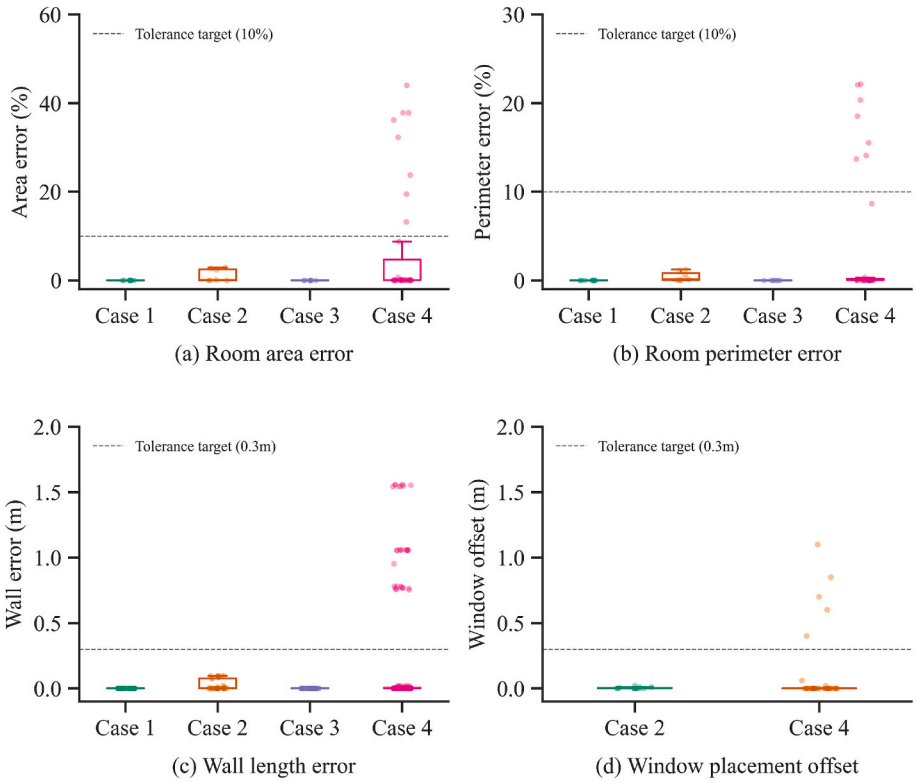


Fig. 17. Boxplot distributions of geometric reconstruction errors across four validation cases.

strictly within the tolerance target lines. This indicates that the system achieves satisfactory reconstruction accuracy in most core functional areas, and deviations in local auxiliary spaces do not compromise the reliability of the model for overall energy prediction. In summary, the quantitative evaluation confirms the robustness and applicability of the proposed method in handling diverse and imperfectly annotated hand-drawn tasks.

3.4.3. *Sensitivity analysis of the regularization parameter λ*

To quantitatively evaluate the influence of the regularization parameter λ on the coordinate inference results, this study selected the most topologically complex multi-story building (Case 4) for sensitivity testing. The experiment set the range of λ between 10^{-2} and 10^{-7}, observing its impact on geometric reconstruction accuracy through a comparison spanning several orders of magnitude. Fig. 18 illustrates the statistical distribution (including boxplots and trend lines) of four metrics, namely room area error, perimeter error, wall length error, and window layout offset, under different λ values.

In the early design stage, the primary concern is whether the accuracy of the majority of rooms and walls is sufficient to support energy simulation, as a few anomalous rooms can be manually corrected later. Therefore, the 75th percentile (Q3) serves as a more representative and robust indicator; it is less affected by extreme values and better reflects the typical performance of the method. The mean value serves as an auxiliary constraint, requiring strict limitation only when it exceeds tolerance thresholds (e.g., area error $>$ 10%). Experimental data indicates that $\lambda = 10^{-4}$ represents a critical turning point for the algorithm's performance. As λ decreases stepwise from 10^{-2} to 10^{-4}, the mean values of various error metrics show an upward trend, whereas the 75th percentile (Q3) continues to decline. This indicates that the errors for the majority of samples (over 75%) significantly decrease, resulting in a more compact and robust distribution. This trend suggests that a larger regularization coefficient exerts an "over-smoothing" effect on coordinate inference. Although strong penalties suppress extreme deviations caused by annotation conflicts (thereby maintaining a lower mean), they interfere with the response of most normal wall segments to physical closure constraints, compromising the accuracy of over 75% of the samples. As λ decreases to 10^{-4}, physical closure constraints gradually regain dominance, leading to a significant improvement in geometric reconstruction accuracy for the vast majority of rooms, as evidenced by the continuous optimization of Q3 metrics. Although a small number of complex areas with sparse data or severe annotation conflicts exhibited larger local deviations due to weakened regularization constraints, which led to a slight rise in the mean, this mean value remains within the acceptable tolerance range for early-stage design.

As λ further decreases to 10^{-7}, the mean and Q3 of all geometric metrics stabilize, and errors no longer fluctuate with parameter changes. This indicates that within this range, the solution to the equation system is dominated entirely by the physical constraints of room closure and pixel scale references. Here, λ serves only to prevent matrix singularity and ensure numerical stability, without interfering with the true physical meaning of the geometric reconstruction. Based on this analysis, this study prioritizes the performance of the majority of samples (focusing on Q3 and the median) while ensuring the mean remains within tolerance. Consequently, $\lambda = 10^{-4}$ was selected as the optimal configuration for the coordinate inference algorithm. This parameter value not only ensures the convergence of the equation solution in the presence of annotation conflicts or sparse data but also effectively minimizes the occurrence of extreme deviations, thereby guaranteeing the engineering reliability of the final generated model.

3.4.4. *Failure modes and user guidance*

Although the proposed method demonstrates good robustness in most scenarios, stress testing on extreme input scenarios identified several typical failure modes (as shown in Fig. 19). These failures primarily stem from the limitations of the image recognition algorithm in handling non-linear features, interference from complex visual elements, and deviations in multimodal semantic understanding. The following is a detailed analysis of these typical failure.

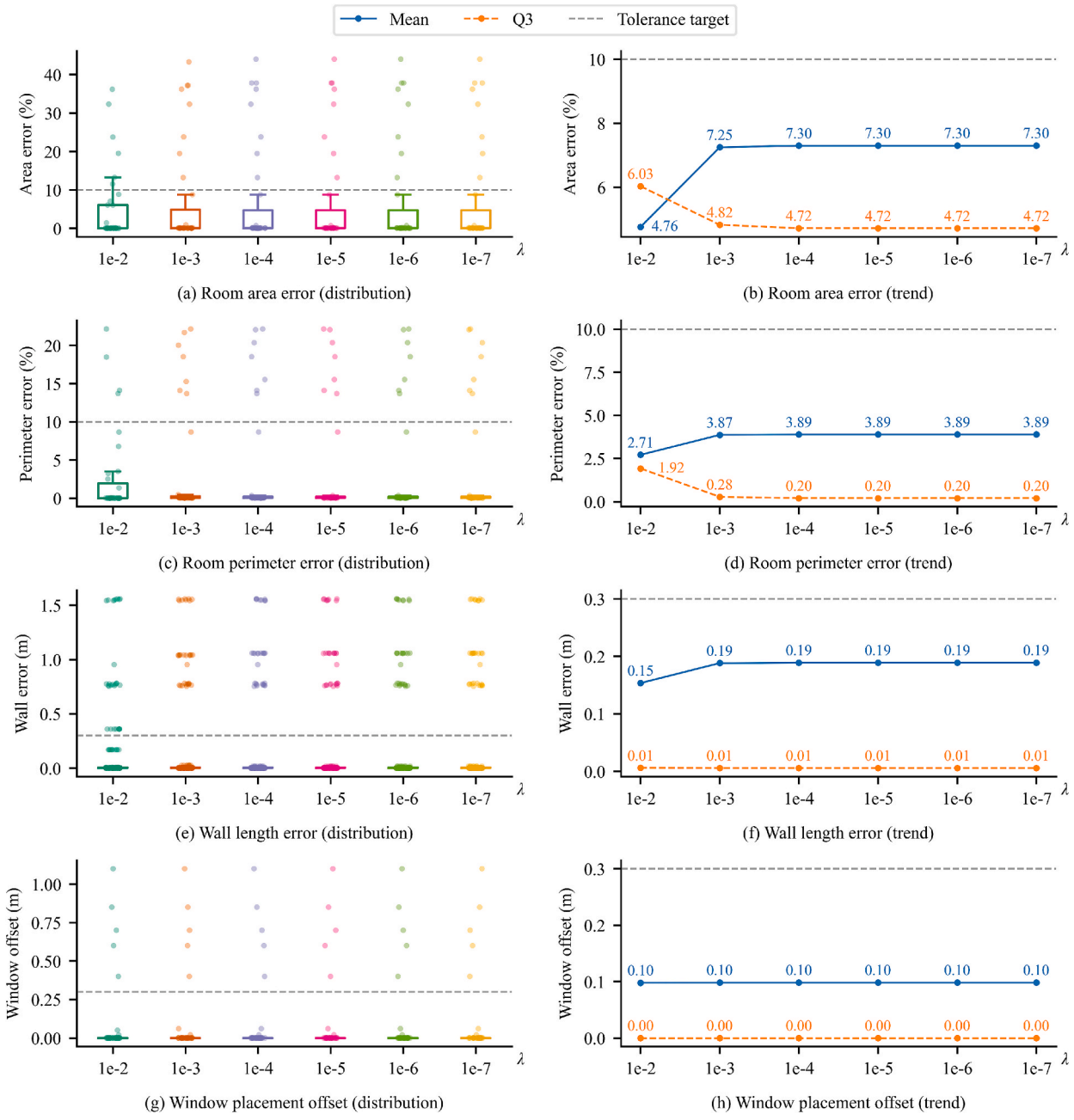
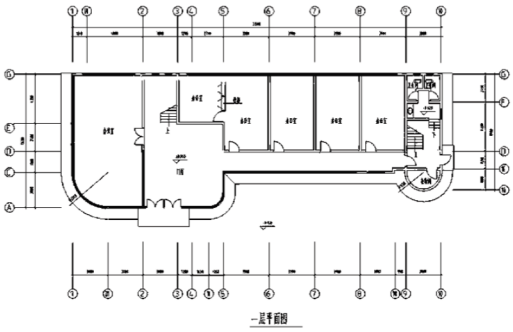
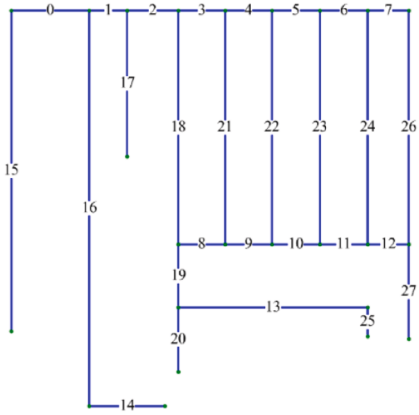


Fig. 18. Sensitivity analysis of the regularization parameter λ on geometric reconstruction accuracy.

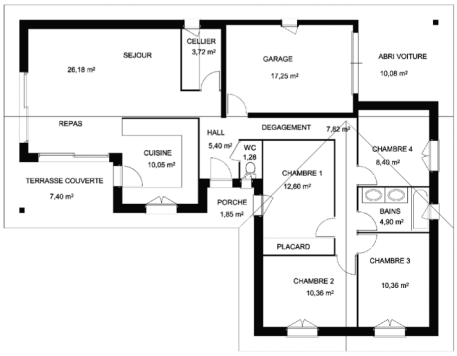
1) **Recognition of curved/non-orthogonal walls and open boundaries**: The current recognition logic prioritizes the extraction of long linear features from the image. As shown in Fig. 19(a), when the floor plan includes curved walls (such as circular balconies or bay windows), the system fails to identify them as continuous architectural boundaries. In the reconstruction result shown in Fig. 19(b), the curved sections are completely lost, retaining only the connected straight segments. Additionally, as shown in Fig. 19(c), the "Cuisine" space failed to be recognized because it was designed as a semi-open area lacking clear solid wall boundaries in the sketch. Since the underlying logic of EnergyPlus requires every Thermal Zone to possess a complete closed loop, this absence of physical walls causes the topology to remain open, preventing the space from being instantiated and ultimately resulting in the room's omission from the energy simulation.
2) **Visual interference from dense annotations and non-architectural lines**: The density of lines in the image significantly impacts recognition accuracy. In the "Toilet" area shown in Fig. 19 (c), the user clustered excessive dimension lines and textual notes within a very small space. These lines overlapped heavily with the wall contours, creating severe visual noise that caused the wall recognition in that area to collapse completely (Fig. 19(d)). Furthermore, auxiliary lines common in architectural plans (such as the roof projection lines in the bottom-right corridor) are often mistaken for solid wall lines. This interference not only prevented the correct recognition of the corridor area but also caused severe geometric distortion.
3) **Multilingual annotations**: Accurate room function recognition is a prerequisite for the automatic assignment of thermal parameters. As shown in Fig. 19(c), when sketch labels use a mix of Chinese and



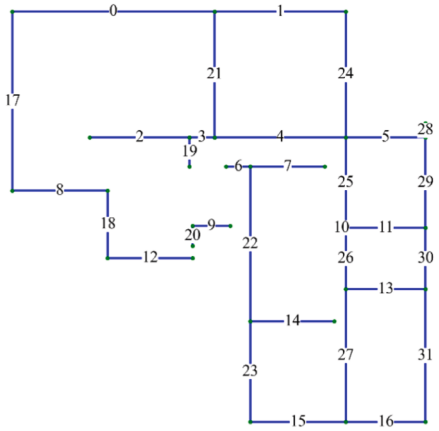
(a) Original sketch with curved and non-orthogonal walls



(b) Reconstruction failure: Missing curved geometry and open boundaries



(c) Original sketch with dense annotations, auxiliary lines, and mixed-language labels



(d) Comprehensive failure: Geometric distortion, wall recognition collapse, and semantic errors

Fig. 19. Examples of typical failure modes: Comparison between hand-drawn sketches and reconstruction results.

English (e.g., "厨房Cuisine "), the multimodal LLM exhibits deviations in language translation and classification mapping.

Based on the aforementioned failure modes, this study provides the following drawing guidelines for users seeking simulation-ready energy models:

1) Linear approximation of curves: Given that arcs cannot currently be processed directly, users are advised to approximate curved contours using multiple short straight segments to ensure the algorithm can successfully capture and close spatial boundaries.
2) Minimizing annotation interference: It is recommended to place dimension annotations outside the building contour. Avoid stacking excessive symbols and text in small spaces (like WCs) to maintain the visual clarity of wall contours.
3) Ensuring physical boundary closure: Ensure that functional zones are closed during drawing to guarantee the successful instantiation of Thermal Zones.
4) Removing interfering auxiliary lines: Sketches used for recognition should ideally retain only walls, doors, windows, and function labels, avoiding complex roof lines or decorative hatching to prevent geometric distortion.

Standardizing function labels: It is recommended to use clear, printed labels (uniformly using English is preferred), and ensure that labels are located in the center of the room to improve the accuracy of recognition and parameter matching.

3.5. *Thermal parameter generator performance evaluation*

This section aims to evaluate the effectiveness of the RAG-based Thermal Parameter Generator (Task 3). The core mechanism of this module is to use the key thermal parameters extracted in Task 1 to construct a weighted query vector. By calculating the weighted Euclidean distance, it retrieves the most physically similar reference model from the model database to generate thermal parameters that meet the modeling requirements.

The weights for the key features were initially set based on the results of sensitivity analyses from relevant literature [63–65], as detailed in Table 6. It is noteworthy that the method allows users to customize these weights based on specific design objectives. For example, in a project located in a high-latitude region, a user could increase the weight of WALLU to prioritize the retrieval of cases with superior insulation performance. This flexibility not only enhances the relevance of the retrieval results but also supports researchers in exploring the impact of specific parameters on energy consumption.

Table 6
Key features and weights for the case study.

Parameter	Value	Weight
LOCATION	Atlanta	0.25
AREA	150 m^2	0.04
NL	1	0.02
CR	0.69	0.1
WWR	0.29	0.15
WALLU	0.75 W/(m^2·K)	0.12
WALLSP	57.6 kJ/(m^2·K)	0.05
ROOFU	0.18 W/(m^2·K)	0.08
WINDOWU	2.02 W/(m^2·K)	0.15
SLABU	1.8 W/(m^2·K)	0.04

Table 7 presents the retrieval results. Since the model database used in this study originates from the U.S. Department of Energy (DOE), its geographical data is confined to the United States, which precluded finding a local case for the target building located in Shanghai. However, the retrieval algorithm transcended this geographical limitation by identifying Atlanta, which shares the same ASHRAE Climate Zone 3A as Shanghai, as the best-matching location. This result demonstrates the algorithm's capability to find a reference model with a highly similar climatic environment for the target building, thereby ensuring the physical plausibility and applicability of the provided contextual information, such as construction layers and material specifications.

After retrieving the most similar case, the LLM generates detailed constructions and materials for the target building by referencing the construction layers of the matched reference model. If the generated parameters, upon verification, do not satisfy the modeling requirements (e.g., the overall wall U-value (WALLU) deviates from the target value), the LLM operates under thermal parameter guardrails that enforce physical constraints and design priorities. It then iteratively refines the parameters, preferentially by adjusting material layer thicknesses or correcting thermal conductivity values, to precisely tune the total thermal resistance until the target performance metrics are met.

To quantify the critical role of the RAG mechanism in this task, this study compared the performance of the proposed method against Prompt-only baseline. The Prompt-only method utilized the exact same structured prompt as the proposed method but intentionally omitted the retrieved similar case as context. Table 8 provides a detailed comparison of the two methods' performance in generating thermal parameters for the exterior wall and exterior window. The analysis clearly reveals the core deficiencies of the Prompt-only method. First, it produces parameters with mismatched names (e.g., a construction layer named "Stucco_1" paired with a material named "Stucco"), which would result in definition errors in EnergyPlus. More critically, while the parameters generated by the Prompt-only method appear reasonable on the surface, their calculated thermal performance (e.g., exterior wall U-value of 0.62 W/(m^2·K)) deviates significantly from the user-defined target (0.75 W/(m^2·K)) specified in Task 1. In contrast, the proposed method, guided by the contextual information provided by RAG, not only generated parameters with physically consistent definitions, but its calculated thermal performance also precisely matched the user's requirements.

Furthermore, the materials generated by the proposed method (e.g., F07 25 mm stucco) not only follow standardized naming conventions but also have thermal conductivity (0.72 W/m·K) and thickness (0.0254 m) values that strictly fall within the permissible ranges specified in standard building material handbooks (e.g., ASHRAE or DOE reference libraries). This result strongly demonstrates that the context provided by RAG acts as a critical physical constraint, effectively preventing the LLM from producing "plausible-but-incorrect" hallucinations under unconstrained conditions and ensuring the accuracy and physical plausibility of the final generated parameters.

Table 7
Top 5 best matching models and similarity scores.

No.	Model name	Score
1	OfficeMedium_Atlanta	0.8323
2	ApartmentMidRise_Atlanta	0.8271
3	ApartmentLowRise_Atlanta	0.8228
4	OutPatientHealthCare_Atlanta	0.7854
5	OfficeSmall_Atlanta	0.7749

3.6. Operational Parameter Generator performance evaluation

This section evaluates the performance of the Operational Parameter Generator (Task 4) in handling incomplete user inputs. EnergyPlus require that all operational schedules (e.g., occupancy, equipment status) cover all hours. However, users typically only define core activity periods in their natural language descriptions (e.g., "occupancy is 80% during working hours"), resulting in incomplete data. The core challenge for this module is to intelligently generate a complete and logical operational schedule based on this limited input.

In the current implementation of the algorithmic logic, if the user does not explicitly specify variations based on date types (e.g., weekdays, weekends, or holidays) in the natural language modeling requirements, the system defaults to a uniform schedule without distinguishing between these date types. Under this setting, the 24-hour hourly operational profiles automatically generated by Task 4 are applied consistently across all 365 days of the year.

To validate the proposed method's advantages in this task, a comparative experiment was once again designed against Prompt-only baseline. The Prompt-only method attempts to directly parse the user's input without any external knowledge. In contrast, the proposed method first uses semantic similarity to retrieve the most relevant, complete reference schedule from the model database that matches the user's requirements (e.g., "Office" function) and injects it into the prompt as context. Table 9 provides a detailed comparison of the outputs from both methods. By leveraging the retrieved similar case, the proposed method successfully understood the typical operational pattern of an office. It not only accurately reproduced the user-defined core working hours but, more importantly, it logically inferred that the occupancy during undefined periods, such as nights and weekends, should be zero. This resulted in a complete, data-sufficient schedule ready for direct use in simulation. The Prompt-only method, however, exposed its limitations in the absence of context. It merely transcribed the information explicitly mentioned in the user's input, returning "null" for all unspecified time periods. Such an output is incomplete and cannot meet the modeling requirements of EnergyPlus without tedious manual completion.

This comparative experiment strongly demonstrates that the RAG mechanism is key to achieving the automated generation of operational parameters. By introducing a validated reference case as context, the proposed method can effectively compensate for the inherent incompleteness of user inputs, significantly enhancing the completeness, plausibility, and direct usability of the generated operational parameters.

3.7. Simulation parameters and boundary conditions

The energy simulations in this study were conducted using the EnergyPlus v9.6.0 engine. All weather data files (EPW) were downloaded from the official EnergyPlus weather database (energyplus.net). Taking the Shanghai region as an example, the selected weather file was CHN_Shanghai.Shanghai.745130_CSWD.epw, which includes hourly dry-bulb temperature, solar radiation, humidity, and other key meteorological elements. For comparative experiments across different climate zones, corresponding standard weather files for representative cities were obtained from the same official source.

Regarding HVAC system modeling, given that the study focuses on the early design stage and emphasizes the influences of building geometry, envelope constructions, and operational schedules, all thermal

Table 8
Comparison of generated thermal parameters by proposed method and prompt-only.

Object	Method	Construction	Material	Calculated feature
External Wall	Proposed method	Outside Layer: F07 25 mm stucco; Layer 2: G01 16 mm gypsum board; Layer 3: Insulation wall; Layer 4: G01 16 mm gypsum board	1. Name: F07 25 mm stucco; Roughness: Smooth; Thickness: 0.0254; Conductivity: 0.72; Density: 1400; Specific Heat: 840; Thermal Absorptance: 0.9; Solar Absorptance: 0.7; Visible Absorptance: 0.7; 2. Name: G01 16 mm gypsum board; Roughness: MediumSmooth; Thickness: 0.0159; Conductivity: 0.16; Density: 800; Specific Heat: 1090; Thermal Absorptance: 0.9; Solar Absorptance: 0.7; Visible Absorptance: 0.7; 3. Name: Insulation wall; Roughness: MediumSmooth; Thermal Resistance:1.09279; Thermal Absorptance: 0.9; Solar Absorptance: 0.7; Visible Absorptance: 0.7;	WALLU: 0.75 W/(m^2·K) WALLSP: 57.6 kJ/(m^2·K)
	Prompt-only	Outside Layer: Stucco_1; Layer 2: Insulation_1; Layer 3: GypsumBoard_1;	1. Name: Stucco; Roughness: Smooth; Thickness: 0.02; Conductivity: 0.69; Density: 1858; Specific Heat: 840; Thermal Absorptance: 0.9; Solar Absorptance: 0.7; Visible Absorptance: 0.7; 2. Name: Insulation_1; Roughness: MediumSmooth; Thermal Resistance: 1.5; Thermal Absorptance: 0.9; Solar Absorptance: 0.5; Visible Absorptance: 0.6; 3. Name: GypsumBoard_1; Roughness: MediumRough; Thickness: 0.0127; Conductivity: 0.16; Density: 785; Specific Heat: 830; Thermal Absorptance: 0.9; Solar Absorptance: 0.5; Visible Absorptance: 0.5;	WALLU: 0.62 W/(m^2·K) 39.49 kJ/(m^2·K)
External Window	Proposed method	Outside Layer: Double Glazing 6 mm	1. U-factor: 2.06; Solar Heat Gain Coefficient: 0.35; Visible Transmittance: 0.55	WINDOWU: 2.26 W/(m^2·K)
	Prompt-only	Outside Layer: Glazing Layer	1. U-factor: 2.06; Solar Heat Gain Coefficient: 0.259; Visible Transmittance: 0.348	WINDOWU: 2.26 W/(m^2·K)

Table 9
Comparison of generated schedules using proposed method and prompt-only.

Method	Retrieved similar schedule (RAG)	Generated schedule
Proposed method	Office_Standard_Atlanta: The occupancy rate starts at 0% until 08:00; rises to 50% at 10:00; peaks at 95% from 11:00 to 12:00; declines to 80% at 13:00 and further to 70% by 14:00; and reaches peak 95% again from 14:00 to 17:00; drops sharply to 30% by 18:00; and returns to 0% at 18:00.	00:00–08:00: 0%, 08:00–09:00: 10%, 09:00–10:00: 50%, 10:00–11:00: 95%, 11:00–13:00: 80%, 13:00–17:00: 95%, 17:00–18:00: 30%, 18:00–24:00: 0%
Prompt-only	–	00:00–08:00: null, 08:00–09:00: 10%, 09:00–10:00: null, 10:00–11:00: 95%, 11:00–12:00: 80%, 12:00–13:00: null, 13:00–17:00: 95%, 17:00–18:00: 30%, 18:00–24:00: null

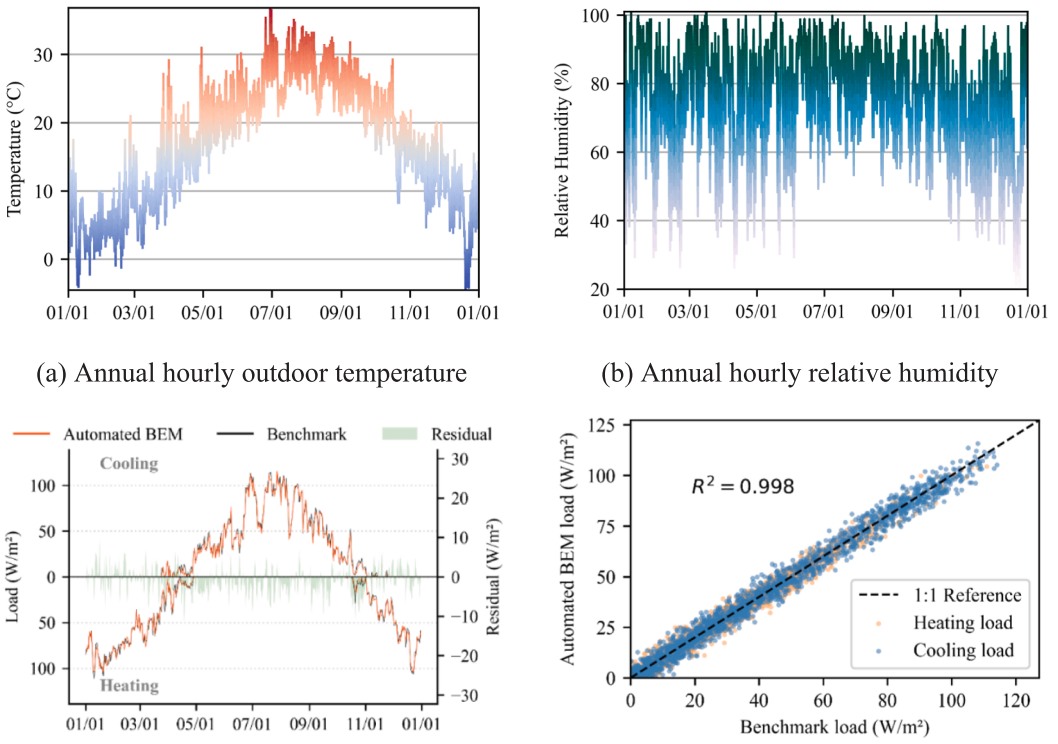
zones were uniformly configured with ZoneHVAC:IdealLoadsAirSystem (ideal loads air system). This system directly computes zone cooling and heating demands, excluding variations arising from the efficiency fluctuations of complex air-conditioning equipment. In compliance with ASHRAE Standard 62.1, infiltration was set to a constant 0.5 air changes per hour (ACH), while outdoor air supply was determined based on occupancy (20 m^3/h per person).

Internal heat gains (including occupant, lighting, and equipment densities) were automatically assigned based on the room function labels identified in Task 2, with values referenced from the performance criteria for office buildings in ASHRAE Standard 90.1. The simulation timestep was set to 4 (corresponding to a 15-minute calculation interval) to balance simulation convergence accuracy with computational efficiency. The heat balance algorithm employed was the Conduction Transfer Function (CTF), with a maximum surface temperature limit of 200°C. The solar distribution algorithm was set to FullExterior to accurately capture the impact of window placement on indoor solar heat gain. Zone Sizing Calculation was enabled for the simulation, with a uniform Sizing Factor of 1.0. Key output variables included the hourly zone ideal load rates (e.g., Zone Ideal Loads Supply Air Total Cooling/Heating Rate) and the total building heating/cooling metrics (DistrictCooling/Heating:Facility). This detailed and standardized documentation of simulation settings aims to provide a clear experimental baseline for other researchers, thereby ensuring the reproducibility and validity of the automated modeling workflow proposed in this study.

3.8. Simulation accuracy and efficiency

This section provides a comprehensive performance evaluation of the final output from the proposed method. To this end, the model generated by proposed method is directly compared against the benchmark model. The benchmark model is a precise representation of the case study building, created in EnergyPlus following standard modeling procedures based on detailed design documents. It includes complete building geometry, detailed envelope constructions, accurate material thermal parameters, and standard operational schedules. This benchmark model serves as the reference standard for assessing proposed method 's modeling performance.

Using Case 1 located in Shanghai (ASHRAE Climate Zone 3A) as a representative example, Fig. 20 comprehensively illustrates the environmental boundary conditions and load simulation results. Fig. 20(a) and (b) present the annual hourly outdoor dry-bulb temperature and relative humidity curves for Shanghai, respectively, reflecting the region's typical characteristics of hot, humid summers and cold, damp winters. Fig. 20(c) compares the hourly cooling and heating load curves of the automated model against those of the benchmark model, while synchronously displaying the prediction residuals between the two. The results indicate that the simulation curves exhibit a high degree of overlap throughout the year, with the residual distribution fluctuating around the zero line. Furthermore, the scatter plots in Fig. 20 (d) reveal that the annual hourly simulation data points cluster closely along the 1:1 diagonal, demonstrating a strong linear correlation. These results confirm that our system possesses reliable load prediction capabilities.



(a) Annual hourly outdoor temperature

(b) Annual hourly relative humidity

(c) Comparison of hourly cooling and heating loads with residual plot

(d) Scatter plot of predicted versus benchmark annual hourly loads

Fig. 20. Comparison of simulated hourly loads for automated generation model and benchmark model under Shanghai weather conditions.

To further quantify the differences in the load results between the two modeling methods, this study employs the Peak Load Error Rate (PLER, Equation (17)) to measure accuracy of the proposed method in predicting the maximum building load. Additionally, the Mean Absolute Error (MAE, Equation (18)), Coefficient of Variance of the Root Mean Square Error (CV(RMSE), Equation (19), and the Coefficient of Determination (R^2, Equation (20)) are used to evaluate the overall consistency and goodness of fit of the hourly load time series from both models over the entire year (8760 h).

$$\text{PLER} = \frac{|y_{max} - \hat{y}_{max}|}{y_{max}} \times 100\% \tag{17}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{18}$$

$$CV(RMSE) = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}}{\bar{y}} \tag{19}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{20}$$

where $\hat{y}_{max}$ denotes the peak load of the model from the proposed method, and y_{max} represents the peak load of the benchmark model. Similarly, y_i is the load of the benchmark model at hour i, and $\hat{y}_i$ is the load of the model from the proposed method at hour i. The term $\bar{y}$ refers to the annual average load of the benchmark model. The total number of hours in a year is denoted by n (typically 8760 h).

Table 10 summarizes the calculated performance metrics for each case. Regarding peak loads, the values predicted by the proposed method align closely with the benchmark, maintaining an average PLER of less than 5%. For annual hourly loads, the average R^2 exceeds 0.995, while the CV(RMSE) remains below 20%. The quantitative analysis results indicate that the proposed method achieves high-accuracy load prediction across diverse building morphologies (including L-shaped, U-shaped, atrium-style, and multi-story bar-shaped layouts) and varying climate zones (e.g., 3A Shanghai/Chongqing, 4A Beijing, and 2A Guangzhou).

Table 10
Simulation accuracy metrics for the model generated by automated generation model the benchmark model.

Case		PLER (%)	CV(RMSE) (%)	R^2
Case 1	Cooling load	1.20	10.83	0.9981
	Heating load	2.86	11.82	0.9978
Case 2	Cooling load	8.33	23.56	0.9915
	Heating load	9.82	25.24	0.9927
Case 3	Cooling load	1.98	8.57	0.9954
	Heating load	0.85	6.95	0.9969
Case 4	Cooling load	6.27	17.66	0.9938
	Heating load	7.56	19.78	0.9943
Mean	–	4.86	15.55	0.9951

The minor discrepancies in the load results primarily stem from the fundamental differences in the construction methods between the automated generation models and the benchmark model. First, the envelope and material properties generated by proposed via RAG inherently differ from the parameters precisely entered into the benchmark model based on design documents. Furthermore, proposed method intelligently infers schedules from incomplete user inputs, whereas the benchmark model directly employs standard, deterministic operational schedules. It is these subtle differences in the generation process that lead to the slight deviations in the final simulation results.

In terms of time efficiency, the proposed method demonstrates a

significant advantage. Table 11 provides a detailed breakdown of the modeling time and API costs for the four validation cases across various stages. The results reveal an upward trend in total time and cost as the complexity of the building morphology increases (e.g., from single-story to multi-story). Specifically, the multi-story bar-shaped building (Case 4) required layer-by-layer processing of geometric topology, resulting in approximately double the time and cost in the geometric generation stage compared to single-story cases, leading to a total time of 12 min and 08 s and a cost of $0.85.

Nevertheless, all cases completed the workflow from sketch input to the generation of a simulation-ready IDF file and the completion of annual load simulation within 15 min, with an average total time of about 8 min. Regarding cost, due to highly optimized prompt structures and token-based billing for API calls, the geometric generation stage (relying on VLM APIs) accounted for nearly the entire expense. The average cost per case was $0.49, as other stages ran locally at no cost. Compared to traditional manual modeling, which typically requires hours or even days, the proposed method demonstrates significant advantages in both efficiency and economic viability, making it particularly suitable for rapid iteration scenarios in early-stage design.

In contrast, traditional manual modeling not only takes several hours or even days but also requires significant expertise and software proficiency from the modeler. By automating tedious steps such as geometric reconstruction, parameter configuration, and file generation, the proposed method effectively shortens the modeling cycle and significantly lowers the barrier to entry for BEM technology.

4. Discussion and future work

4.1. Comparison with existing early-stage BEM tools

The “Sketch-to-Energy” workflow proposed in this study differs in positioning from existing early-stage Building Energy Modeling tools, including BIM/gbXML-based workflows (e.g., OpenStudio/SketchUp + gbXML importers), parametric design plugins (e.g., Rhino/Grasshopper + Ladybug/Honeybee), and recently developed deep learning-based floor plan recognition methods (e.g., DeepFloorplan). Table 12 provides a comprehensive comparison across multiple dimensions.

As shown in Table 12, traditional BIM or gbXML-based tools rely heavily on precise modeling inputs and are typically suitable for the detailed design stage. While these tools can map partial parameters through predefined templates, geometric errors and missing parameters often require significant manual correction. Parametric platforms, although highly flexible, still require professional scripting skills for geometric construction and parameter configuration. Although recent deep learning methods have achieved high automation in wall extraction, they exhibit poor robustness to noisy sketches, lack deep semantic interpretation of room functions, and are incapable of automatically configuring physical dimensions or thermal parameters, thus necessitating tedious post-processing.

In contrast, by integrating multimodal LLMs with computer vision techniques, the proposed method achieves the construction of complete EnergyPlus models from floor plan sketches and natural language description, significantly lowering the modeling barrier and reducing manual intervention. However, manual correction remains a crucial step for ensuring model reliability. In complex failure scenarios involving non-orthogonal walls or dense annotations (as detailed in Section 3.4.3), users are still required to perform certain sketch optimizations or fine-tune the generated geometric model. This method does not aim to replace existing professional BIM-to-BEM workflows; rather, it serves as a rapid pre-processing step to convert sketches into simulation models, supporting subsequent refined design in tools like Rhino or OpenStudio.

4.2. Core advantages of the proposed method

The method proposed in this study integrates multimodal LLMs, computer vision techniques, and RAG technology, aiming to achieve automatic generation and simulation of EnergyPlus models from hand-drawn architecture sketches and natural language modeling requirements. The method significantly streamlines the BEM process by enhancing efficiency, lowering technical barriers, and ensuring reliability of load predictions, offering an end-to-end intelligent solution for the building performance simulation domain. Its core advantages can be summarized in four key points.

First, the method supports a combination of sketches and text, overcoming the limitations of purely text-based descriptions. Compared to a unimodal text input, this multimodal approach conveys a building's geometric features and spatial layout more intuitively and precisely. This effectively prevents geometric modeling errors arising from ambiguous or misinterpreted textual descriptions, thereby enhancing both the convenience and accuracy of user input.

Second, the method generates geometric models with robustness and adaptability. The method leverages advanced multimodal LLMs (e.g., GPT-4o) and computer vision models to generate structured 3D building geometry from hand-drawn sketches. Traditional rule-based methods are highly dependent on the regularity and completeness of drawings. Vision models like DeepFloorplan, when applied to hand-drawn sketches, are sensitive to noise and often fail by misinterpreting dimensional annotations as walls or being unable to extract room functions. Conversely, relying solely on LLMs to directly generate vector graphics often introduces geometric distortions that compromise the original spatial topology. The proposed method effectively integrates these technologies through a task decomposition strategy. The multimodal LLMs perform high-level semantic understanding and logical reasoning to identify the overall spatial layout, distinguish valid geometric elements from noise, and parse textual annotations. This is followed by a vision model that executes pixel-level geometric segmentation on the cleaned data. The advantages of our method in handling complex hand-drawn sketches has been verified.

Third, the method uses RAG to enhance model domain knowledge for generating thermal and operational parameters. It utilizes RAG to inject rich and context-specific information into a locally deployed lightweight model (LLaMA-3-8B-Instruct), enabling it to generate highly relevant and physically plausible building thermal parameters and operational schedules based on the building type and local climate. Experimental results confirm that, compared to a prompt-only method lacking an external knowledge base, the application of RAG effectively prevents parameter mismatches and the omission of critical information,

Table 11
Breakdown of time and costs for the automated modeling workflow across the four typical validation cases.

Case	Task 1	Task 2	Task 3	Task 4	Task 5	Total time	Total cost
Case 1	15/0.00	160/0.37	83/0.00	50/0.00	72/0.00	6:20	0.37
Case 2	19/0.00	185/0.42	92/0.00	67/0.00	87/0.00	7:31	0.43
Case 3	15/0.00	152/0.31	89/0.00	48/0.00	55/0.00	5:59	0.31
Case 4	19/0.00	437/0.85	96/0.00	70/0.00	106/0.00	12:08	0.85
Mean	17/0.00	234/0.49	90/0.00	59/0.00	80/0.00	8:00	0.49

Note: Data presented as time (s) / cost ($). Total time is in min:s; Costs primarily originate from multimodal LLM API calls (GPT-4o and Gemini-2.5-Pro) during the geometric generation stage; other stages utilize local LLaMA-3-8B and local EnergyPlus execution, incurring no additional fees.

Table 12
Comprehensive comparison of the proposed method with existing BEM workflows.

Category	Input format	Geometric automation	Parameters automation	Modeling cost	Manual correction	Application Stage
BIM/gbXML (e.g., OpenStudio)	Precise BIM model / gbXML	Semi-automatic	Semi-automatic	High	High (all geometric errors and parameters)	Middle to late-stage
Parametric Tools (e.g., Rhino/ Honeybee)	Parametric geometry scripts	Semi-automatic	Semi-automatic	Medium	Medium (Script debugging, parameter definition)	Design iteration
Deep Learning Methods (e.g., DeepFloorplan)	Clean floor plans	High	Low	High	High (Semantic annotation, hyper-parameter settings)	Early-stage
Proposed Method (Sketch-to-Energy)	Floor plan sketches + Natural language	High	High	Low	Low (Manual required for failure modes)	Early-stage

demonstrating enhanced consistency and completeness in parameter generation.

Finally, concerning operational cost and efficiency, the method achieves optimal allocation of computational resources. As noted in related research [46], "prompt-only" methods often require larger, more advanced models like GPT-4o to ensure generation quality, which can incur API call costs of several dollars for a single small office building, even for text-only tasks. Proposed method optimizes this by selectively invoking high-performance APIs only for the most critical and complex task of geometry parsing (with an average cost of $0.37 across all case studies), while leveraging a cost-effective, locally deployed open-source model (LLaMA-3-8B-Instruct on a single NVIDIA RTX 3090 GPU) for text-based parameter generation. Compared to workflows that rely exclusively on large models like GPT-4o for all tasks, this strategy effectively reduces the operational cost per model and minimizes dependence on expensive APIs, achieving an optimal balance between high-quality output and cost-effectiveness.

As for the data privacy and ethical handling problem, the system is designed to support the local deployment of lightweight VLMs and inference engines, enabling users to process sensitive design data entirely within a secure, offline environment. Furthermore, for workflows utilizing commercial cloud-based APIs (e.g., GPT-4o), the system strictly adheres to data privacy protocols by configuring settings to exclude user data from model training. The uploaded sketches and prompts are not retained or utilized to improve the foundational models.

In summary, the proposed method provides architects and non-expert users with an efficient and low-cost pathway to BEM, significantly advancing the intelligent integration of building design and energy optimization. Its primary application scenarios include:

1) Rapid energy modeling in early design stages. Quickly generate energy models from preliminary sketches and key textual descriptions to provide immediate feedback on design modifications, enabling designers to compare the energy performance of different alternatives (e.g., layouts, WWR) in real-time.

2) Energy analysis for non-experts. By enabling intuitive interaction through sketches and natural language, the method substantially lowers the technical barrier to BEM, empowering users from non-technical backgrounds to easily conduct building energy analyses.

3) Large-scale building energy assessment. When combined with automated batch processing scripts, the method can rapidly generate models based on common geometric information and features (e.g., WWR, U-values), making it suitable for energy assessment projects involving a large number of buildings.

4.3. Limitations and future work

While the proposed method demonstrates significant potential for automating BEM, its widespread practical application requires addressing key limitations in three core areas: the scope of geometric processing, dependency on the knowledge base, and the level of modeling granularity.

First, the geometric processing capability of the current method remains limited by input complexity and format. It currently excels at processing single-story, hand-drawn floor plans. However, it lacks the capability to handle the complex 3D spatial relationships of multi-story buildings (e.g., stairwells, atria) and cannot directly parse detailed, layered CAD drawings (e.g., DWG files). Furthermore, its ability to reconstruct complex geometric forms (such as sloped roofs and irregular facades) is limited, constraining its direct application in non-standard and architecturally complex projects. To overcome these barriers, future work will extend the geometric reconstruction module to support vector-based CAD parsing, enabling the direct interpretation of complex engineering drawings. Concurrently, we will explore the local deployment of lightweight Vision-Language Models (VLMs) to reduce reliance on costly closed-source APIs, thereby addressing data privacy and operational cost concerns.

Second, the accuracy of parameter generation is highly dependent on the completeness of the RAG knowledge base. The method's performance is strongly correlated with the quality and coverage of the data in its retrieval library. If the knowledge base lacks energy models for specific climate zones, atypical building types (e.g., data centers, laboratories), or novel construction materials, the RAG system may fail to provide accurate context. This can lead to biased or incomplete parameter generation, thereby compromising the reliability of the final simulation results. Future work will aim to build a more comprehensive knowledge base by incorporating a wider variety of cases, thereby enhancing the generalization capability and precision of parameter generation.

Third, there is a lack of modeling granularity. The current method tends to apply global assumptions during parameter assignment, for instance, by defaulting to a single envelope construction for all exterior walls. This makes it difficult to satisfy fine-grained design requirements, such as assigning different constructions and thermal properties to facades with varying orientations (e.g., a south-facing glass curtain wall versus a north-facing insulated wall). Future efforts will employ advanced prompt engineering to enhance the system's ability to parse and execute complex instructions, enabling differentiated parameter assignment for specific components (e.g., individual walls or windows on specific floors).

Additionally, to support goal-driven design for green certification, future work will integrate optimization algorithms (e.g., genetic algorithms or reinforcement learning). By setting performance metrics from green building standards (e.g., LEED) as targets, the system will automatically adjust key design parameters (e.g., material selection) until simulation results meet requirements, thereby providing designers with optimized solutions that satisfy performance goals. Through these enhancements, proposed method is poised to evolve into a more intelligent, goal-driven BEM tool, better equipped to address diverse climate conditions, complex architectural designs, and green certification challenges, thereby providing robust technical support for the sustainable development of the architecture industry.

5. Conclusion

This study proposes an automated BEM method designed to address the core challenges of traditional BEM, including inefficiency, high

technical barriers, and strict data formatting requirements. By integrating multimodal LLMs, computer vision techniques, and RAG, proposed method enables the end-to-end automation of generating standardized EnergyPlus models from hand-drawn floor plans and textual requirements. The method decouples the BEM process into five core tasks: requirement parsing, geometric reconstruction, thermal parameter generation, operational parameter generation, and simulation execution, thereby systematically enhancing the workflow's automation and intelligence. Requirement parsing (Task 1) achieves accurate interpretation of natural language. Geometric reconstruction (Task 2), powered by multimodal LLMs and computer vision techniques, effectively resolves the noise and non-standard nature of hand-drawn sketches. RAG-driven parameter generation (Tasks 3 and 4) ensures the physical plausibility and completeness of both thermal and operational inputs. The overall efficiency of the workflow is further enhanced through automated parameter integration, mapping, and simulation processes via Python scripts (Task 5). The case study results demonstrate that proposed method exhibits advantages in modeling efficiency, load prediction accuracy, system usability, and operation cost.

1) Modeling efficiency. Proposed method automates the entire workflow from sketch and text to a simulation-ready EnergyPlus model. For the four tested cases, the average total time is about 8 min. This represents a substantial time saving compared to the laborious configuration and debugging inherent in manual modeling, with efficiency gains being particularly pronounced for complex, multi-zone buildings.

2) Prediction accuracy. In the validation across different climate zones (2A, 3A, 4A) and diverse building morphologies (L-shape, U-shape, Courtyard, and Multi-story Bar), the method exhibits strong predictive consistency. All cases achieved an average PLER below 5%, an average R^2 value for annual hourly load exceeding 0.995, and an average CV(RMSE) below 20% against the benchmark models. This performance is attributed to a multi-stage information processing pipeline. In Task 1, structured prompting ensures 99.4% accuracy in requirement parsing, providing reliable inputs for subsequent tasks. In Task 2, a key innovation is the synergistic use of multimodal LLMs (GPT-4o, Gemini-2.5-Pro) and a specialized vision model (DeepFloorplan). This hybrid approach robustly handles inherent imperfections in hand-drawn sketches (such as line noise, discontinuities, and scale distortions) to accurately reconstruct complex geometry, achieving a mean wall length error of 0.05 m and a room function classification accuracy of 91.2%. It demonstrates enhanced robustness compared to using LLMs alone to generate vector graphics (which often introduces geometric errors) or relying solely on vision models (which fail to process textual annotations and noise). In Tasks 3 and 4, RAG retrieves relevant parameters from a historical BEM library, ensuring physical realism and data completeness while mitigating LLM hallucinations.

3) System usability and design support. Proposed method significantly lowers the technical barrier to entry for BEM. By accepting multimodal inputs (sketches and text), the method translates unstructured design intent into machine-readable data (JSON) and a final simulation file (IDF). This process avoids the ambiguities of purely text-based descriptions and empowers users to initiate simulations without mastering complex BEM software. It offers non-experts an intuitive "Design-to-Analysis" workflow. A key contribution of this method lies in its support for design decision-making. As highlighted in the retrieval mechanism, the method allows for the flexible configuration of feature weights. This empowers designers to perform sensitivity analyses on specific performance indicators, thereby identifying the most effective strategies for energy optimization during the conceptual phase.

4) Operation cost. Economic viability is achieved through a hybrid-inference strategy. Proposed method selectively invokes powerful but costly commercial APIs (GPT-4o, Gemini-2.5-Pro) only for the most demanding visual understanding task (geometry parsing). For text-centric tasks like parameter generation, it utilizes a locally deployed, lightweight open-source model (LLaMA-3-8B-Instruct on a single NVIDIA GeForce RTX 3090 GPU). The average generation cost per case is only $0.49. This approach dramatically reduces operational costs and reliance on expensive APIs compared to workflows dependent exclusively on large commercial models or extensive manual labor, ensuring a scalable and economically feasible solution.

However, the proposed method has several limitations. First, the scope of geometric processing is limited. The system cannot directly parse complex CAD drawings, such as DWG files. Additionally, manual correction is still required in specific failure modes, such as sketches with mixed-language annotations, dense interference lines, or open boundaries. Second, the accuracy of parameter generation relies heavily on the completeness of the RAG knowledge base. Deviations may occur if the database lacks specific atypical building types or novel construction materials. Third, the current modeling granularity is insufficient. It is difficult to achieve differentiated parameter assignment at the specific building components.

Overall, this study provides an automated bridge connecting early architectural design with energy performance simulation, enabling the rapid conversion of conceptual sketches into executable models for energy analysis. Although the validation results presented here reflect performance based on the four specific cases demonstrated, the proposed method exhibits significant potential for broader application across diverse architectural contexts.

CRediT authorship contribution statement

Yake Zhang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guansong Wang:** Writing – review & editing, Visualization, Validation, Supervision, Formal analysis. **Peng Xu:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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